The five theorems of modular Koszul duality

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Abstract

Koszul duality over a point terminates: a quadratic algebra determines a dual coalgebra, the bar construction mediates between them, and no further structure appears. Over an algebraic curve, the theory does not terminate. The bar differential acquires curvature from the Hodge bundle on moduli space; its failure to square to zero is controlled by a single scalar $\kappa(\mathcal{A})$, the modular characteristic, extracted from the collision residue of the universal Maurer–Cartan element.

We prove five structural theorems for the bar complex of a chirally Koszul chiral algebra $\mathcal{A}$ on a smooth algebraic curve X. Theorem A constructs the bar-cobar adjunction and its Verdier intertwining, producing from the bar coalgebra $\overline{B}_X(\mathcal{A}) = T^c(s^{-1}\bar{A})$ the homotopy Koszul dual $\mathcal{A}^!_\infty$. Theorem B proves bar-cobar inversion $\Omega(\overline{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ on the Koszul locus: unconditionally at genus 0 and on the standard landscape at all genera. Theorem C decomposes the genus-g ambient complex into Verdier-Lagrangian complements $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) \cong H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$, with the Koszul complementarity sum $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$ family-dependent: 0 for Kac–Moody and free fields, 13 for Virasoro. Theorem D identifies the genus expansion: the genus-1 formula $\mathrm{obs}_1(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_1$ is unconditional for every family. On the uniform-weight lane, the all-genera extension $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ (UNIFORM-WEIGHT) holds via clutching morphisms on $\partial\overline{\mathcal{M}}_g$, with generating function the $\hat{A}$-genus; the clutching-uniqueness step is the principal remaining conditional element (an independent GRR derivation provides an alternative path, §6.4). Theorem H proves that chiral Hochschild cohomology is concentrated in degrees $\{0, 1, 2\}$ with a polynomial Hilbert series controlled by the centre and its dual.

All five theorems are Σ_n-coinvariant projections of a single E_1 object: the ordered bar complex $\overline{B}^{\mathrm{ord}}_X(\mathcal{A})$ with its deconcatenation coproduct and classical R-matrix $r_{\mathcal{A}}(z)$, for which $\mathrm{av}(r_{\mathcal{A}}(z)) = \kappa(\mathcal{A})$ for abelian algebras; for non-abelian Kac–Moody, $\kappa = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ after the Sugawara shift. The Heisenberg algebra provides the complete base computation; every other family is a variation with increasingly complex collision residues.

Contents

1 Introduction: from a point to a curve

1.1 The problem

Koszul duality over a point terminates. A quadratic algebra $A = T(V)/(R)$ determines a dual coalgebra $A^! = (H^*(B(A)))^\vee$, the bar construction $B(A) = (T^c(s^{-1}\bar{A}), d_B)$ mediates between them, and the cobar functor inverts: $\Omega(B(A)) \xrightarrow{\sim} A$ when A is Koszul [9, 2, 6]. The bar cohomology concentrates in degree 1, the derived categories of A-modules and $A^!$-modules are equivalent, and no further structure appears.

Over an algebraic curve, three obstructions prevent this from working. The generators of a chiral algebra are sections of a $\mathcal{D}_X$-module, not elements of a vector space. The relations are operator product expansions with poles of arbitrary order, not quadratic expressions. The topology of higher-genus curves forces the fiberwise bar differential to satisfy $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ rather than $d^2 = 0$, where $\omega_g = c_1(\mathbb{E})$ is the Hodge class on $\overline{\mathcal{M}}_g$.

The question is: what structure replaces the terminating classical theory?

1.2 The answer

The ordered bar complex

$$\overline{B}_X^{\mathrm{ord}}(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}), \qquad |s^{-1}v| = |v| - 1, \tag{1}$$

with deconcatenation coproduct

$$\Delta[s^{-1}a_1|\cdots|s^{-1}a_n] \;=\; \sum_{i=0}^{n}[s^{-1}a_1|\cdots|s^{-1}a_i] \otimes [s^{-1}a_{i+1}|\cdots|s^{-1}a_n], \tag{2}$$

is a cofree E_1-coalgebra. Here $\bar{\mathcal{A}} = \ker(\varepsilon\colon \mathcal{A} \to \omega_X)$ is the augmentation ideal and s^{-1} is the desuspension, lowering degree by 1. The integral kernel is the logarithmic 1-form

$$\eta_{12} \;=\; d\log(z_1 - z_2) \;=\; \frac{dz_1 - dz_2}{z_1 - z_2}\,, \tag{3}$$

with conformally invariant residue 1: no coordinate, no metric, no level. At three points, the Arnold relation

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} \;=\; 0 \tag{4}$$

generates all relations in $H^*(\mathrm{Conf}_n(\mathbf{C}), \mathbf{C})$. The bar differential squares to zero if and only if the n-th products satisfy the Borcherds identity; the Arnold relation kills the differential form, the Borcherds identity kills the algebra, and both are consumed simultaneously. This is the *selection principle*: configuration space geometry determines the algebraic axioms; the bar construction accepts exactly chiral algebras.

The binary collision residue of the bar differential is the classical R-matrix

$$r_{\mathcal{A}}(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}) \;\in\; \mathrm{End}_{\mathcal{A}}(2)[\![z^{-1}]\!],$$

where $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is the Maurer–Cartan element of the bar differential, satisfying $d\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ automatically because $(D_{\mathcal{A}})^2 = 0$. The Σ_n-coinvariant quotient erases the matrix structure and retains only the scalar

$$\kappa(\mathcal{A}) \;=\; \mathrm{av}(r_{\mathcal{A}}(z)) \quad \text{for abelian and scalar families,} \tag{5}$$

while for non-abelian affine Kac–Moody

$$\kappa(V_k(\mathfrak{g})) \;=\; \mathrm{av}(r_{\mathcal{A}}(z)) + \dim(\mathfrak{g})/2.$$

the modular characteristic: the single number that survives averaging. Five structural theorems extract the Σ_n-invariant content of the ordered bar. Each theorem characterizes one facet of $\Theta_{\mathcal{A}}$:

(A) *Arena*: $\Theta_{\mathcal{A}}$ exists (bar-cobar adjunction and Verdier intertwining).

(B) *Faithfulness*: $\Theta_{\mathcal{A}}$ determines $\mathcal{A}$ (bar-cobar inversion on the Koszul locus).

(C) *Decomposition*: $\Theta_{\mathcal{A}}$ splits (Lagrangian complementarity under Verdier involution).

(D) *Leading coefficient*: $\Theta_{\mathcal{A}}$ has universal scalar projection ($\mathrm{obs}_1 = \kappa \cdot \lambda_1$ unconditional; $\mathrm{obs}_g = \kappa \cdot \lambda_g$ at all genera on the uniform-weight lane via clutching/GRR).

(E) *Coefficient ring*: $\Theta_{\mathcal{A}}$ has polynomial coefficients ($\mathrm{ChirHoch}^*$ concentrated in $\{0, 1, 2\}$).

The chain is forced: without A, there is no arena for the dual; without B, the bar transform can lose the boundary algebra; without C, higher-genus deformation theory has no canonical polarization; without D, the genus tower has no distinguished scalar shadow; without H, the coefficient ring is uncontrolled.

1.3 Prior work

Five existing programmes address aspects of the problem. Beilinson–Drinfeld [1] construct factorization algebras on $\mathrm{Ran}(X)$ and prove chiral homology is well-defined, but do not treat bar-cobar inversion, complementarity, or the genus tower. Francis–Gaitsgory [4] establish chiral Koszul duality at genus 0 for Lie-type bar complexes; their framework does not reach higher genera or the full tensor bar. Costello–Gwilliam [3] develop factorization algebras for perturbative QFT; they do not treat Koszul duality or the modular characteristic. Gui–Li–Zeng [5] prove a quadratic duality at the cochain level, a genus-0 degree-2 result. Malikov–Schechtman [8] construct homotopy chiral algebras as the correct input, but do not prove the five structural theorems.

Programme	**A** adjunction	**B** inversion	**C** compl.	**D** genus	**H** chiral Hochschild
BD [1]	✓				
FG [4]	✓	✓*			
CG [3]					
GLZ [5]	✓*				
MS [8]					
This work	✓	✓	✓	✓	✓

*FG: genus 0, Lie-type. GLZ: genus 0, degree 2.

1.4 The Heisenberg computation

The Heisenberg chiral algebra $\mathcal{H}_k$ is generated by a single current J of conformal weight 1 with OPE $J(z)\,J(w) \sim k/(z-w)^2$. One generator, one double pole, no simple pole. The binary residue of the bar differential is the complete computation:

$$d_{\overline{B}}[s^{-1}J \,|\, s^{-1}J] \;=\; \mathrm{Res}_{z_1=z_2}\left[\frac{k}{(z_1-z_2)^2}\cdot d\log(z_1-z_2)\right] \;=\; k.$$

The modular characteristic is $\kappa(\mathcal{H}_k)=k$, the collision residue $r(z)=k/z$ (scalar, not matrix), bar-cobar inversion gives $\Omega(\overline{B}(\mathcal{H}_k))\simeq\mathcal{H}_k$, complementarity yields $\kappa(\mathcal{H}_k)+\kappa(\mathcal{H}_k^!)=k+(-k)=0$, genus universality produces $F_1=k/24$ and the $\hat{A}$-series at all genera (UNIFORM-WEIGHT), and chiral Hochschild gives $P(t)=1+t+t^2$. Every clause of every theorem is verified in a single example. Every other family is a variation on this computation, with increasingly complex collision residues.

1.5 Conventions

Throughout, X denotes a smooth projective curve over $\mathbf{C}$, $\mathcal{A}$ a chiral algebra on X in the sense of Beilinson–Drinfeld [1], and $\overline{\mathcal{M}}_g=\overline{\mathcal{M}}_{g,0}$ the Deligne–Mumford moduli space of stable curves of genus g. Grading is cohomological: $|d|=+1$. Desuspension lowers degree: $|s^{-1}v|=|v|-1$. The bar complex uses the augmentation ideal: $\overline{B}(\mathcal{A})=T^c(s^{-1}\bar{\mathcal{A}})$ with $\bar{\mathcal{A}}=\ker(\varepsilon)$. The Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n(X)$ is the real oriented blowup of X^n along all diagonal strata; it is a manifold with corners whose codimension-one faces D_S are indexed by subsets $S\subseteq\{1,\ldots,n\}$ with $|S|\geq 2$.

2 The bar complex on algebraic curves

2.1 Chiral algebras

A *chiral algebra* $\mathcal{A}$ on a smooth curve X [1] is a $\mathcal{D}_X$-module equipped with a chiral bracket

$$\mu^{\mathrm{ch}}\colon j_*j^*(\mathcal{A}\boxtimes\mathcal{A})\longrightarrow\Delta_!(\mathcal{A}), \tag{6}$$

where $j\colon X^2 \setminus \Delta \hookrightarrow X^2$ is the complement of the diagonal and $\Delta_!$ is the !-pushforward along the diagonal embedding. In local coordinates, the chiral bracket encodes the operator product expansion:

$$a(z)\, b(w) \;\sim\; \sum_{n\geq 0} \frac{a_{(n)}b(w)}{(z-w)^{n+1}}.$$

The zeroth product $a_{(0)}b$ is the Lie bracket (first-order pole); the first product $a_{(1)}b$ carries the symmetric inner product (second-order pole). The Borcherds identity

$$\sum_{j\geq 0} \binom{m}{j} (a_{(n+j)}b)_{(m+k-j)}c \;=\; \sum_{j\geq 0} (-1)^j \binom{n}{j} \big(a_{(m+n-j)}(b_{(k+j)}c) - (-1)^n\, b_{(n+k-j)}(a_{(m+j)}c)\big) \tag{7}$$

is the defining axiom: it couples all pole orders in a single constraint.

2.2 The propagator and the Arnold relation

The bar differential extracts OPE data from collisions on $C_2(X) = X \times X \setminus \Delta$. A regular form has residue zero and extracts nothing. A double pole transforms under coordinate change and has no coordinate-invariant residue. A simple pole is forced; among 1-forms with a first-order pole along Δ, only the logarithmic derivative has conformally invariant residue:

$$\eta_{12} = d\log(z_1 - z_2) = \frac{dz_1 - dz_2}{z_1 - z_2}\,, \qquad \mathrm{Res}_\Delta\ \eta_{12} = 1.$$

This is the *propagator*: weight 1, no dependence on level, metric, or conformal structure.

Under $z \mapsto w(z)$, the propagator transforms as $\eta_{12} \mapsto \eta_{12} + d\log \frac{w(z_1)-w(z_2)}{z_1-z_2}$; near the diagonal the correction reduces to $d\log w'(z_i)$, the standard cocycle for $\mathrm{Diff}(X)$. This failure of invariance is not a defect; it is the datum that the Virasoro algebra measures.

At three points, the Arnold relation

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$$

restates the partial-fractions identity

$$\frac{1}{(z_1-z_2)(z_2-z_3)} + \frac{1}{(z_2-z_3)(z_3-z_1)} + \frac{1}{(z_3-z_1)(z_1-z_2)} = 0.$$

Arnold proved that, replicated across all triples $\{i,j,k\} \subset \{1,\dots,n\}$, this generates the complete ideal of relations in $H^*(\mathrm{Conf}_n(\mathbf{C}), \mathbf{C})$.

2.3 The ordered bar complex

Definition 2.1 (Ordered bar complex)**.** Let $\mathcal{A}$ be a chiral algebra on a smooth curve X, $\bar{\mathcal{A}} = \ker(\varepsilon)$ the augmentation ideal. The *ordered bar complex* is

$$\overline{B}^{\mathrm{ord}}_X(\mathcal{A}) \;=\; \bigoplus_{n\geq 1} (s^{-1}\bar{\mathcal{A}})^{\otimes n} \otimes \Gamma\big(\overline{\mathrm{FM}}_n(X),\, \Omega^{n-1}_{\log}\big), \tag{8}$$

with no Σ_n-quotient. Here s^{-1} denotes desuspension ($|s^{-1}a| = |a| - 1$), and the cohomological convention $|d| = +1$ is used throughout.

The bar differential decomposes into three pieces:

$$d_{\overline{B}} = d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}}. \tag{9}$$

The internal differential d_{int} acts on each tensor factor of $\mathcal{A}$. The de Rham differential d_{dR} acts on logarithmic forms on $\overline{\mathrm{FM}}_n(X)$. The residue differential d_{res} extracts OPE data at collision divisors: for a codimension-one face $D_{\{i,j\}}$ corresponding to the collision $z_i \to z_j$,

$$d_{\mathrm{res}}^{(ij)}\big(a_1 \otimes \cdots \otimes a_n \otimes \eta_{ij} \wedge \omega'\big) \;=\; \pm\, a_1 \otimes \cdots \otimes \widehat{a_i} \otimes \cdots \otimes \widehat{a_j} \otimes \cdots \otimes a_{i(p)}a_j \otimes \omega', \tag{10}$$

where p is the pole order, hats denote omission, and $a_{(p)}b$ is the p-th Fourier mode of the OPE.

2.4 Why $d^2 = 0$: the Borcherds selection principle

Theorem 2.2 (Bar nilpotence)**.** *The bar differential satisfies $d_{\overline{B}}^2 = 0$ if and only if the n-th products of $\mathcal{A}$ satisfy the Borcherds identity* (7)*.*

Proof sketch. Expand $(d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}})^2$ into nine terms. Three vanish individually: $d_{\mathrm{int}}^2 = 0$ (cochain complex), $d_{\mathrm{dR}}^2 = 0$ (de Rham), and $\{d_{\mathrm{int}}, d_{\mathrm{dR}}\} = 0$ (Leibniz). The remaining terms are controlled by index overlap in the residue operator:

Disjoint indices ($\{i,j\} \cap \{k,\ell\} = \emptyset$): residue operators commute; cancellation is by the Koszul sign rule. No condition on the OPE.

One shared index (say $j = k$): three boundary strata meet at the codimension-2 locus $D_{ij\ell}$ of triple collision. The cyclic sum of iterated residues vanishes if and only if the Borcherds identity holds. The Arnold relation kills the differential form; the Borcherds identity kills the algebra. Both are consumed.

Same pair ($\{i,j\} = \{k,\ell\}$): $(\mathrm{Res}_{D_{ij}})^2 = 0$ by degree. □

Remark 2.3 (Selection table). The bar construction asks the same question at every level of geometry:

Base	**Condition for $d^2 = 0$**	**Selected objects**
Point	Jacobi identity	Lie algebras
Curve X	Borcherds identity	Chiral algebras
X over $\overline{\mathcal{M}}_g$	+ period corrections	Modular Koszul algebras

Each row is forced by the geometry of the next.

2.5 Two coproducts

The ordered bar carries the *deconcatenation coproduct* (2): $n+1$ terms, coassociative but not cocommutative, an E_1-coalgebra reflecting the linear ordering. The Σ_n-coinvariant quotient

$$\overline{B}{}^{\Sigma}_X(\mathcal{A}) \;=\; \bigoplus_{n \geq 1} (s^{-1}\bar{\mathcal{A}})^{\otimes n}_{\Sigma_n} \otimes \Gamma\big(\overline{\mathrm{FM}}_n(X),\, \Omega^{n-1}_{\log}\big)$$

carries the *coshuffle coproduct*: 2^n terms, coassociative and cocommutative, an E_∞-coalgebra. The ordered bar is the primitive object; the symmetric bar is the lossy projection that discards the R-matrix.

2.6 From genus 0 to all genera

At genus $g \geq 1$, the function $z_1 - z_2$ has no global meaning; the propagator must be replaced by $d \log E(z_1, z_2)$, where $E(z, w)$ is the prime form on a Riemann surface of genus g (a section of $K^{-1/2} \boxtimes K^{-1/2}$, weight 1). The bar differential acquires curvature:

$$(d_{\text{fib}}^{\text{ord}})^2 \;=\; r_{\mathcal{A}}(z) \cdot \omega_g, \tag{11}$$

matrix-valued in $\operatorname{End}_{\mathcal{A}}(2) \otimes H^{1,1}(\mathcal{C}_g)$, where $\omega_g = c_1(\mathbb{E})$ is the Hodge class on $\overline{\mathcal{M}}_g$. Averaging over Σ_2 collapses this to the scalar equation $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$.

The *total* bar differential $D_{\mathcal{A}} = \sum_{g \geq 0} \hbar^g \, d_{\mathcal{A}}^{(g)}$ absorbs the curvature into the genus expansion: $D_{\mathcal{A}}^2 = 0$ unconditionally.

Definition 2.4 (Modular characteristic)**.** The *modular characteristic* of a chiral algebra $\mathcal{A}$ is

$$\kappa(\mathcal{A}) := \operatorname{av}(r_{\mathcal{A}}(z)),$$

the Σ_2-coinvariant of the collision residue r-matrix.

Definition 2.5 (Maurer–Cartan element)**.** The *Maurer–Cartan element* is $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$, the positive-genus part of the bar differential. Since $D_{\mathcal{A}}^2 = 0$, it satisfies

$$d_0 \, \Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0 \tag{12}$$

unconditionally. Existence of the MC element is a tautology of the bar construction; the five theorems characterize its structure.

3 Theorem A: bar-cobar adjunction and Verdier intertwining

3.1 Statement

The bar complex produces a coalgebra; the question is what algebra it encodes. Three functors act on the bar complex and produce three distinct outputs:

$$\begin{array}{lcll} \Omega(\overline{B}(\mathcal{A})) & \simeq & \mathcal{A} & \text{(inversion: recovers the original)}, \\ \mathbb{D}_{\text{Ran}}(\overline{B}(\mathcal{A})) & \simeq & \mathcal{A}^!_\infty & \text{(Verdier: the homotopy dual)}, \\ (H^*(\overline{B}(\mathcal{A})))^\vee & = & \mathcal{A}^! & \text{(linear duality: the strict dual)}. \end{array} \tag{13}$$

Confusing any two is a category error: cobar inverts, Verdier dualizes at chain level, and linear duality produces the strict dual.

Definition 3.1 (The homotopy Koszul dual $\mathcal{A}^!_\infty$; explicit construction)**.** For an augmented chiral algebra $\mathcal{A}$ on a smooth curve X over a field of characteristic zero, define the *homotopy Koszul dual factorization algebra* by

$$\mathcal{A}^!_\infty \;:=\; \mathbb{D}_{\text{Ran}}\big(\overline{B}_X(\mathcal{A})\big),$$

the Verdier dual of the bar factorization coalgebra on $\operatorname{Ran}(X)$. Equivalently (Lowen–Positselski homotopy-coalgebra duality for conilpotent complete objects), $\mathcal{A}^!_\infty$ is the chiral algebra freely generated by $s\,\bar{\mathcal{A}}^\vee$ together with an A_∞-structure obtained by transferring the cooperations of

$\overline{B}_X(\mathcal{A})$ through a contraction onto its cohomology, under the finiteness hypothesis that $\overline{B}_X(\mathcal{A})$ is degreewise finite-dimensional in each bar-depth stratum (satisfied by every freely strongly generated chiral algebra, including Heisenberg, affine Kac–Moody at generic level, Virasoro, and the principal $\mathcal{W}_N$). On the Koszul locus this A_∞-structure strictifies to a genuine chiral algebra structure whose cohomology is the strict Koszul dual: $H^*(\mathcal{A}^!_\infty) \cong \mathcal{A}^!$.

Theorem 3.2 (Theorem A: bar-cobar adjunction, Verdier intertwining, Koszul-conditional inversion)**.** *Let $\mathcal{A}$ be an augmented chiral algebra on a smooth curve X with PBW-filtered bar complex $\overline{B}_X(\mathcal{A})$ in the finite-bar class of Definition* 3.1*.*

(T1a) Bar–cobar adjunction (unconditional in the weight-completed ambient; ambient-qualified otherwise)*: the bar and cobar functors form an adjoint pair*

$$\overline{B}_X \colon \mathrm{Alg}_{\mathrm{aug}}(\mathrm{Fact}(X)) \rightleftarrows \mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X)) \ \colon\! \Omega_X$$

with unit $\eta\colon \mathrm{id} \to \Omega_X \circ \overline{B}_X$ and counit $\varepsilon\colon \overline{B}_X \circ \Omega_X \to \mathrm{id}$, defined on every augmented chiral algebra without any Koszul hypothesis and natural in $\mathcal{A}$ within the Francis–Gaitsgory factorization ambient of [4], provided that ambient is taken in its canonical weight-completed */ pro-conilpotent form* pro-$\mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X))$ *whenever $\mathcal{A}$ has unbounded conformal-weight grading (classes* G, L *with positive-energy grading and, a fortiori, class* M*). On the raw bounded direct-sum ambient* $\mathrm{Ch}(\mathrm{Vect})$ *the target* $\mathrm{CoAlg}_{\mathrm{conil}}$ *is too small for class* M*: the bar coalgebra $\overline{B}(\mathrm{Vir}_c)$ fails conilpotency because $L_0 \in \overline{B}^1$ has $\bar{\Delta}^{(N)}(L_0) \neq 0$ for every $N \geq 1$ (Remark* 3.4*; cf. the same L_0-obstruction underlying Theorem B (B.2),* `thm:chiral-positselski-weight-completed`*).*

(T1b) Verdier intertwining (unconditional, given finiteness)*: Verdier duality on* $\mathrm{Ran}(X)$ *intertwines the bar with the homotopy Koszul dual of Definition* 3.1*:*

$$\mathbb{D}_{\mathrm{Ran}}\, \overline{B}_X(\mathcal{A}) \ \simeq\ \mathcal{A}^!_\infty.$$

No Koszul hypothesis is invoked; $\mathcal{A}^!_\infty$ carries A_∞-structure measuring the failure of strict Koszulness.

(T1c) Koszul-conditional inversion*: if $\mathcal{A}$ is chirally Koszul — i.e. a twisting morphism $\tau\colon \mathcal{C} \to \mathcal{A}$ with $\mathcal{C}$ conilpotent makes $K^L_\tau(\mathcal{A}, \mathcal{C})$ acyclic, or equivalently the PBW spectral sequence collapses at E_2 — then the adjunction of* (T1a) *and the Verdier intertwining of* (T1b) *become quasi-isomorphisms:*

$$\mathcal{C} \xrightarrow{\sim} \overline{B}_X(\mathcal{A}), \qquad \Omega_X(\mathcal{C}) \xrightarrow{\sim} \mathcal{A}, \qquad H^*(\mathcal{A}^!_\infty) \cong \mathcal{A}^!.$$

Only this statement invokes the Koszul hypothesis; (T1a) *and* (T1b) *stand without it.*

Remark 3.3 (Three independent statements, three disjoint inputs)*.* The partition into (T1a), (T1b), (T1c) separates three distinct universal properties with disjoint load-bearing inputs: adjunction uses only the cofree/free universal properties and the Francis–Gaitsgory ambient [4] (weight-completed whenever the conformal-weight grading is unbounded); Verdier intertwining uses the six-functor formalism on $\mathrm{Ran}(X)$ together with the finiteness hypothesis of Definition 3.1; Koszul-conditional inversion uses additional Koszul acyclicity. Bundling them into a single statement conflates three distinct universal properties and obscures which hypothesis each clause requires.

Remark 3.4 (Ambient qualifier for (T1a); class-M at the raw direct sum). The adjunction $\overline{B}_X \dashv \Omega_X$ is a formal consequence of the free/cofree universal properties in the chiral factorization setting of Francis–Gaitsgory [4], *provided* the target coalgebra category genuinely contains $\overline{B}_X(\mathcal{A})$. For class G (Heisenberg) and class L (affine Kac–Moody away from critical level) the bar coalgebra is conilpotent already at the bounded direct-sum ambient Ch(Vect), so the adjunction lands in $\mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X))$ without further completion. For class M (Virasoro, $\mathcal{W}_N$, critical-level Kac–Moody) and, more generally, for any chiral algebra whose bar filtration $F^n = \bigoplus_{k\leq n} B_k$ has unbounded conformal weight, conilpotency of $\overline{B}_X(\mathcal{A})$ fails on the raw direct sum: the L_0-witness $\bar{\Delta}^{(N)}(L_0) \neq 0$ for every $N \geq 1$ in $\overline{B}^1(\mathrm{Vir}_c)$ (chapters/theory/theorem$_{B}scope_{p}latonic.tex, Virrawbarprop$
$conilpotent/weight-completedfactorizationcategory$pro-$\mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X))$ against the strong filtration, in which conilpotency is restored at every finite-weight truncation and the Mittag–Leffler–Milnor passage to the inverse limit converges, exactly as in Theorem B (B.2), `thm:chiral-positselski-w`
`thm:completed-bar-cobar-strong`. In this precise sense (T1a) and Theorem B (B.2) share a single ambient discipline: the unconditional statement is at the weight-completed level, and the raw direct-sum statement for class M is genuinely false (not a gap) and should not be asserted at that ambient.

3.2 Proof sketch

Step 1: The twisting morphism. A chiral twisting morphism $\tau\colon \mathcal{C} \to \mathcal{A}$ is a degree-+1 element in the chiral convolution algebra satisfying the Maurer–Cartan equation $d\tau + \tau \star \tau = 0$, where the star product is mediated by the integration kernel η_{12} on $X^2 \setminus \Delta$. The acyclicity of the twisted tensor product $K^L_\tau(\mathcal{A}, \mathcal{C})$ is the defining property of a chiral Koszul pair.

Step 2: The PBW recognition criterion. The PBW filtration on $\mathcal{A}$ by bar-length makes the associated graded quadratic-Koszul in the classical sense. The chiral algebra need not be quadratic (the Virasoro OPE has a quartic pole, $\mathcal{W}$-algebras have poles of arbitrary order); what matters is that the leading-order structure under the PBW filtration is. The E_2 page of the bar spectral sequence identifies with the classical bar complex of the associated graded, and the collapse at E_2 is the chiral Koszul condition.

Step 3: Verdier intertwining. Verdier duality $\mathbb{D}_{\mathrm{Ran}}$ on $\mathrm{Ran}(X)$ converts coalgebras to algebras. Applied to $\overline{B}_X(\mathcal{A})$, it produces a factorization algebra $\mathcal{A}^!_\infty$ on $\mathrm{Ran}(X)$. The key identity is

$$\mathbb{D}_{\mathrm{Ran}}(T^c(s^{-1}\bar{\mathcal{A}})) \;\simeq\; T(s\,(H^*(\overline{B}(\mathcal{A})))^\vee),$$

where the left side is the Verdier dual of the cofree coalgebra and the right side is the free algebra on the linear dual of bar cohomology. On the Koszul locus, bar cohomology concentrates in degree 1, and the right side reduces to the strict Koszul dual algebra $\mathcal{A}^!$.

Step 4: Filtered comparison. The filtered comparison lemma establishes that the PBW spectral sequence is compatible with the Verdier functor: the filtration on $\overline{B}_X(\mathcal{A})$ induced by bar degree is taken by $\mathbb{D}_{\mathrm{Ran}}$ to the dual filtration on $\mathcal{A}^!_\infty$. This is the non-formal content of the theorem; the compatibility is not automatic and requires the Fulton–MacPherson boundary structure.

Example 3.5 (Heisenberg)**.** For $\mathcal{A} = \mathcal{H}_k$: the bar complex $\overline{B}(\mathcal{H}_k) = T^c(s^{-1}\mathbf{C}\cdot J)$ has bar cohomology concentrated in degree 1 ($H^{-1}(\overline{B}(\mathcal{H}_k)) = \mathbf{C}\cdot s^{-1}J$). The Koszul dual coalgebra is $\mathcal{H}_k^{\mathrm{i}} = \mathbf{C}\cdot s^{-1}J$; the strict Koszul dual is the curved symmetric chiral algebra $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ with curvature $m_0 = -k\omega$. The Verdier dual recovers $\mathcal{H}_k^!$ at the level of cohomology. The complementarity sum: $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = k + (-k) = 0$.

4 Theorem B: bar-cobar inversion on the Koszul locus

4.1 Statement

Theorem A constructs the dual algebra $\mathcal{A}^!_\infty$, but does not prove that the bar transform is faithful: the coalgebra $\overline{B}(\mathcal{A})$ could in principle fail to reconstruct $\mathcal{A}$.

Theorem 4.1 (Theorem B: bar-cobar inversion)**.** *Let $\mathcal{A}$ be a chirally Koszul chiral algebra on a smooth curve X.*

(i) Genus 0 (unconditional). *The cobar-bar counit is a quasi-isomorphism:*

$$\varepsilon\colon \Omega_X(\overline{B}_X(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}.$$

The bar spectral sequence collapses at E_2, and bar cohomology concentrates in bar degree 1: $H^*(B(\mathcal{A})) \cong (s\,\mathcal{A}^{\mathrm{i}})$, identifying the Koszul dual coalgebra.

(ii) Genus $g \geq 1$ (on the modular Koszul lane). *At each genus g, if $\mathcal{A}$ satisfies the diagonal Ext vanishing condition on the fibers of the universal curve over $\overline{\mathcal{M}}_{g,n}$, then the genus-$g$ cobar-bar counit is a quasi-isomorphism: $\Omega^{(g)}(\overline{B}^{(g)}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}^{(g)}$. For the standard landscape (Heisenberg, affine Kac–Moody at generic level, Virasoro at generic central charge, principal $\mathcal{W}$-algebras, lattice vertex algebras), the Ext vanishing holds unconditionally, and bar-cobar inversion is unconditional at all genera.*

4.2 The genus-0 argument

The genus-0 proof proceeds by spectral sequence. The PBW filtration on $\mathcal{A}$ by conformal weight induces a decreasing filtration F^p on $\overline{B}_X(\mathcal{A})$ by bar degree: elements of F^p have at least p tensor factors. The associated spectral sequence has E_1 page equal to the classical bar complex $B(\operatorname{gr}\mathcal{A})$ of the associated graded, and converges to $H^*(\overline{B}_X(\mathcal{A}))$.

The Koszul condition is that $\operatorname{gr}\mathcal{A}$ is quadratic Koszul in the classical sense (Priddy [9]). Then the E_1 differential has cohomology concentrated on the diagonal: $E_2^{p,q} = 0$ for $p \neq q$. The spectral sequence collapses at E_2, and bar cohomology concentrates in bar degree 1. The counit map $\varepsilon\colon \Omega_X(\overline{B}_X(\mathcal{A})) \to \mathcal{A}$ is then a quasi-isomorphism by the comparison theorem for filtered complexes.

4.3 The higher-genus extension

At genus $g \geq 1$, the fiberwise bar differential satisfies $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g \neq 0$: the bar complex is curved, and the cobar functor cannot be applied in the classical sense. The resolution uses the total bar differential $D_{\mathcal{A}}$ with $D^2_{\mathcal{A}} = 0$: the curvature is absorbed into the genus expansion.

The genus-g inversion requires two conditions:

1. *Central curvature*: the curvature $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_g$ is a scalar multiple of the Hodge class, not an arbitrary endomorphism-valued class.

2. *Diagonal Ext vanishing*: the twisted tensor product $K^L_\tau(\mathcal{A}^{(g)}, \mathcal{C}^{(g)})$ is acyclic at genus g.

For the standard landscape, central curvature is automatic (the Virasoro-type curvature is generated by the stress tensor) and diagonal Ext vanishing follows from the PBW argument adapted to the genus-g bar complex. The exception is the $\beta\gamma$ system at integer spin, where the Ext group has an anomalous contribution from the fermionic parity.

Example 4.2 (Heisenberg)**.** $\overline{B}(\mathcal{H}_k)$ has bar cohomology concentrated in bar degree 1: $H^{-1}(\overline{B}(\mathcal{H}_k)) = \mathbf{C} \cdot s^{-1}J$. The cobar recovers $\mathcal{H}_k$ as a graded algebra. Bar-cobar inversion is unconditional at all genera: the curvature $d_{\mathrm{fib}}^2 = k \cdot \omega_g$ is scalar, and the Ext vanishing is automatic because the only OPE pole is of order 2 (no Lie bracket, no higher obstructions).

Remark 4.3 (Off the Koszul locus)*.* Off the Koszul locus, bar-cobar inversion fails: the spectral sequence does not collapse at E_2, bar cohomology is not concentrated in degree 1, and the counit is not a quasi-isomorphism. The failure is measured by the shadow obstruction tower $\{S_r(\mathcal{A})\}_{r\geq 2}$: the degree-$r$ shadow class $S_r \in H^*(\overline{\mathcal{M}}_{0,r})$ is the obstruction to extending the inversion from bar degree $r-1$ to bar degree r. All standard chiral algebras in the landscape are chirally Koszul; the shadow tower measures the *complexity* of their higher-genus behaviour, not the failure of duality itself.

5 Theorem C: complementarity

5.1 The problem

At genus $g \geq 1$, the bar complex of $\mathcal{A}$ and the bar complex of its Koszul dual $\mathcal{A}^!$ coexist on the same moduli space $\overline{\mathcal{M}}_g$. The curvatures are related: $d_{\mathrm{fib}}^2(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \omega_g$ and $d_{\mathrm{fib}}^2(\mathcal{A}^!) = \kappa(\mathcal{A}^!) \cdot \omega_g$. The Verdier involution exchanges the two bar complexes. The question is whether this exchange induces a canonical decomposition of the ambient cohomology.

5.2 Statement

Theorem 5.1 (Theorem C: Lagrangian complementarity)**.** *Let $(\mathcal{A}, \mathcal{A}^!)$ be a chirally Koszul pair.*

(C1) Fiber-center decomposition (unconditional in the coderived category). *For every $g \geq 0$, the genus-g ambient complex $\mathbf{C}_g(\mathcal{A}) := R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ admits a homotopy decomposition*

$$\mathbf{C}_g(\mathcal{A}) \;\simeq\; \mathbf{Q}_g(\mathcal{A}) \;\oplus\; \mathbf{Q}_g(\mathcal{A}^!), \tag{14}$$

where $\mathbf{Q}_g(\mathcal{A})$ and $\mathbf{Q}_g(\mathcal{A}^!)$ are Lagrangian summands with respect to the Verdier pairing.

(C2) Lagrangian eigenspace decomposition (proved for $g \geq 1$). *At the cohomological level,*

$$Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) \;\cong\; H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}}). \tag{15}$$

The two summands are eigenspaces of the Verdier involution, and the decomposition is canonical.

(C3) Shifted-symplectic upgrade (conditional). *If the bar-cobar normal complex is perfect and nondegenerate, the ambient complex carries a $(2-d)$-shifted symplectic structure, and each Lagrangian summand carries a shifted Lagrangian structure. This upgrade is conditional on perfectness and nondegeneracy.*

5.3 The Koszul complementarity sum

The Lagrangian decomposition constrains the modular characteristics of $\mathcal{A}$ and $\mathcal{A}^!$:

$$\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) \;=\; K(\mathcal{A}),$$

where $K(\mathcal{A})$ is the *Koszul complementarity conductor*, a family-dependent constant. This is not a universal zero.

Proposition 5.2 (Complementarity sums)**.** (i) Heisenberg: $\kappa(\mathcal{H}_k) = k,\ \kappa(\mathcal{H}_k^!) = -k,\ K = 0$.

(ii) Affine Kac–Moody: $\kappa(V_k(\mathfrak{g})) = \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee},\ \kappa(V_k(\mathfrak{g})^!) = -\frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee},\ K = 0$.

(iii) Virasoro: $\kappa(\mathrm{Vir}_c) = c/2,\ \kappa(\mathrm{Vir}_{26-c}) = (26-c)/2,\ K = 13$. *Self-dual at $c = 13$.*

(iv) Principal $\mathcal{W}_N$: $\kappa(\mathcal{W}_N) = c \cdot (H_N - 1)$ *where* $H_N = \sum_{j=1}^N 1/j$. *K is nonzero and N-dependent.*

(v) Free fields *(lattice, $\beta\gamma$, bc): $K = 0$.*

Proof sketch. The Lagrangian decomposition of Theorem C forces $\kappa + \kappa^! = \varrho_{\mathcal{A}} \cdot K$ at the level of the scalar shadow, with $\varrho_{\mathcal{A}}$ family-specific ($\varrho_{\mathrm{KM}} = \varrho_{\mathrm{free}} = 0$, $\varrho_{\mathrm{Vir}} = 1/2$, $\varrho_{\mathcal{W}_N} = H_N - 1$, $\varrho_{\mathrm{BP}} = 1/6$). For Heisenberg and affine Kac–Moody, the Koszul dual has opposite modular characteristic (the Verdier involution reverses the sign of κ), giving $K = 0$. For the Virasoro algebra, $c \mapsto 26-c$ under Koszul duality (the ghost central charge), and $K = c/2+(26-c)/2 = 13$, constant. The self-dual point $c = 13$ is the unique value where $\kappa = \kappa' = 13/2$: the Lagrangian decomposition is symmetric. □

Remark 5.3 (Five objects). The five objects appearing in the complementarity theorem: $\mathcal{A}$ (the algebra), $\overline{B}(\mathcal{A})$ (the bar coalgebra), $\mathcal{A}^{\mathrm{i}} = H^*(\overline{B}(\mathcal{A}))$ (the dual coalgebra), $\mathcal{A}^! = (\mathcal{A}^{\mathrm{i}})^\vee$ (the dual algebra), and $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = R\operatorname{Hom}_{\mathcal{A}\otimes\mathcal{A}^{\mathrm{op}}}(\mathcal{A},\mathcal{A})$ (the derived chiral center). These are distinct objects produced by distinct functors; conflation of any two is a category error.

6 Theorem D: the genus expansion

6.1 Statement

Complementarity supplies an ambient polarization of the genus tower, but the scalar projection κ is already bar-intrinsic and must still be identified genus by genus.

Theorem 6.1 (Theorem D: genus universality)**.** *Let $\mathcal{A}$ be a chirally Koszul chiral algebra with modular characteristic $\kappa(\mathcal{A})$.*

(i) Genus 1 (unconditional). *The genus-*1 *obstruction class is*

$$\mathrm{obs}_1(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_1 \in H^*(\overline{\mathcal{M}}_{1,1}),$$

*and the genus-*1 *free energy is $F_1(\mathcal{A}) = \kappa(\mathcal{A})/24$. This holds for every family.*

(ii) All genera, uniform weight (UNIFORM-WEIGHT). *For chiral algebras whose strong generators have a single conformal weight, the obstruction class at every genus is*

$$\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g \quad (g \geq 1),$$

and the free energies are $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \hat{A}_g$, where $\hat{A}_g$ is the g-th coefficient of the $\hat{A}$-genus:

$$\sum_{g\geq 1} F_g(\mathcal{A}) \cdot \hbar^{2g} \;=\; \kappa(\mathcal{A}) \cdot \big(\hat{A}(i\hbar) - 1\big). \tag{16}$$

(iii) Multi-weight correction (ALL-WEIGHT $+\delta F_g^{\mathrm{cross}}$). *For multi-weight algebras at genus $g \geq 2$, the diagonal term remains universal but the full free energy is*

$$F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}(\mathcal{A}),$$

with $\delta F_g^{\mathrm{cross}}$ the cross-channel correction from mixed conformal-weight interactions.

Remark 6.2 (Scope of the all-genera statement). Part (i) is unconditional: the genus-1 formula $\mathrm{obs}_1 = \kappa \cdot \lambda_1$ holds for every chirally Koszul algebra, with no uniform-weight hypothesis and no clutching input. Part (ii) extends the scalar factorization from genus 1 to all genera by propagation along the clutching morphisms on $\partial\overline{\mathcal{M}}_g$. The non-trivial step is *clutching-uniqueness*: the assertion that the scalar factorization propagated through clutching agrees, at each genus, with the direct bar-complex computation. This step uses the bar-intrinsic scalar trace of $\Theta_{\mathcal{A}}$ together with the boundary structure of $\overline{\mathcal{M}}_g$; it is proved on the uniform-weight lane where the graph-sum formula factors through κ alone. An independent derivation via the Grothendieck–Riemann–Roch theorem on the universal curve (§6.4) provides a second path that bypasses clutching-uniqueness entirely. The genus-1 result and the two proof paths (clutching and GRR) are the principal structural elements; the all-genera statement on the uniform-weight lane rests on either one.

6.2 The $\hat{A}$-genus and Bernoulli numbers

The $\hat{A}$-genus has the explicit expansion

$$\hat{A}(x) - 1 \;=\; \frac{x/2}{\sin(x/2)} - 1 \;=\; \frac{1}{24}\,x^2 + \frac{7}{5760}\,x^4 + \frac{31}{967680}\,x^6 + \cdots$$

The coefficients are $\hat{A}_g = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$, where B_{2g} is the $2g$-th Bernoulli number. The leading term $\hat{A}_1 = 1/24$ is universal: for every family, $F_1 = \kappa/24$.

6.3 The shadow obstruction tower

The Maurer–Cartan element $\Theta_{\mathcal{A}}$ projects at each degree $r \geq 2$ to a shadow invariant $S_r(\mathcal{A})$; the tower $\{S_r\}_{r\geq 2}$ classifies chiral algebras into four complexity classes:

Class	**Depth**	Δ	**Example**	**Behaviour**
G (Gaussian)	2	$0\ (S_4 = 0)$	Heisenberg	Formal, free
L (Lie)	3	$0\ (\kappa = 0)$	Aff. KM (crit.)	Single Massey
C (Contact)	4	special	$\beta\gamma$	Contact then rigid
M (Mixed)	∞	$\neq 0$	Virasoro, $\mathcal{W}_N$	Non-formal

The critical discriminant $\Delta = 8\kappa S_4$ governs a dichotomy: $\Delta = 0$ terminates the tower, $\Delta \neq 0$ makes it infinite. The *shadow depth* $r_{\max}$ (the highest degree at which $S_r \neq 0$) takes values $\{2, 3, 4, \infty\}$; the *algebraic depth* $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ measures the internal complexity of the MC equation and admits no value 3 or any finite value ≥ 3: there is a gap.

6.4 Proof architecture

The proof of genus universality proceeds in three steps.

Step 1: Genus 1 *(unconditional).* On $\overline{\mathcal{M}}_{1,1}$, the self-loop graph Γ_{irr} contributes $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_1$, and integration over $\overline{\mathcal{M}}_{1,1}$ gives $F_1 = \kappa/24$. The only invariant that contributes at genus 1 is κ itself (degree 2 of the MC element); all higher shadow coefficients S_r with $r \geq 3$ are invisible at genus 1.

Step 2: Uniform-weight factorization. For uniform-weight algebras, the scalar obstruction class at every genus equals $\kappa(\mathcal{A}) \cdot \lambda_g$ (Proposition 6.4). Two independent proofs are given there: (A) via Grothendieck–Riemann–Roch on the universal curve, identifying the bar curvature with the Hodge bundle curvature through the Arakelov–Faltings theorem; (B) via clutching propagation from genus 1, using the Whitney sum formula and the injectivity of the trace on the one-dimensional tautological line.

Step 3: Multi-weight correction. When generators have distinct conformal weights, cross-channel propagators produce mixed terms that do not factor through κ alone. The correction $\delta F_g^{\mathrm{cross}}$ is explicit and computable from the full R-matrix data of the ordered bar.

Remark 6.3 (Genus-1 anchor). Step 1 is unconditional: $\mathrm{obs}_1 = \kappa \cdot \lambda_1$ holds for every chirally Koszul algebra, with no uniform-weight hypothesis and no clutching input. The genus-1 result of Theorem D(i) is independent of the all-genera mechanism.

The passage from genus 1 to all genera is the content of the following proposition, which closes the all-genera extension by two independent routes.

Proposition 6.4 (Scalar obstruction equals Hodge Euler class)**.** *Let $\mathcal{A}$ be a chirally Koszul chiral algebra on the Koszul locus whose strong generators all have the same conformal weight (the uniform-weight hypothesis). Let $\pi \colon \overline{\mathcal{C}}_g \to \overline{\mathcal{M}}_g$ be the universal curve, $\mathbb{E} = R^0\pi_*\omega_\pi$ the Hodge bundle of rank g, and $\lambda_g = c_g(\mathbb{E})$ its top Chern class. Then for all $g \geq 1$:*

$$\mathrm{obs}_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g \qquad (UNIFORM-WEIGHT). \tag{17}$$

Proof. Two independent proofs are given. Each uses the genus-1 identification $\mathrm{obs}_1 = \kappa \cdot \lambda_1$ as base case and extends to all genera by a different mechanism.

Lemma 6.5 (Uniform-weight scalar factorization of the bar complex; shared input to Proofs A and B)**.** *On the uniform-weight lane, the bar complex of a modular Koszul chiral algebra $\mathcal{A}$ over a fixed smooth projective curve Σ_g factorizes as an external tensor product*

$$\overline{B}(\mathcal{A})\big|_{\Sigma_g} \;\simeq\; \overline{B}^{\mathrm{scalar}}(\mathcal{A}) \otimes_{\mathbb{Q}} \Omega^{\bullet}_{\Sigma_g}(\log D)\big|_{\text{unif-wt}},$$

with $\overline{B}^{\mathrm{scalar}}(\mathcal{A})$ a one-dimensional algebraic factor carrying the curvature scalar $\kappa(\mathcal{A})$ (the modular characteristic), and $\Omega^{\bullet}_{\Sigma_g}(\log D)$ the logarithmic de Rham complex with log-poles on the collision divisor $D \subset \Sigma_g^{\times n}$. As the curve varies over $\overline{\mathcal{M}}_g$, the zero-mode sector of the *geometric factor is the direct image $R^0\pi_*\omega_\pi = \mathbb{E}$ (the Hodge bundle), and the scalar virtual class in $K^0(\overline{\mathcal{M}}_g) \otimes \mathbb{Q}$ is*

$$[\overline{B}^{(g)}_{\mathrm{scalar}}(\mathcal{A})]^{\mathrm{vir}} \;=\; \kappa(\mathcal{A}) \cdot [\mathbb{E}].$$

Proof. The uniform-weight hypothesis forces every generator of $\mathcal{A}$ to have the same conformal weight, so every OPE propagator reduces to a scalar multiple of the single weight-1 form $d\log E(z,w)$ (the prime form differential). The bar differential is therefore a tensor of an algebraic piece (scalar channel, recording the Casimir coefficient) with a purely geometric piece (log-de-Rham). The zero-mode reduction from the full $\Omega^\bullet_{\Sigma_g}$ to $R^0\pi_*\omega_\pi = \mathbb{E}$ is the standard Hodge-theoretic restriction at the constant term. The virtual class equation then follows because the scalar channel contributes the constant $\kappa(\mathcal{A})$ while the geometric channel contributes $[\mathbb{E}]$. □

Remark 6.6 (Two proofs, one input). Lemma 6.5 is the common load-bearing input to the two proofs of Theorem T4 that follow. Proof A derives $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ from this input by Grothendieck–Riemann–Roch plus the Arakelov–Faltings identification of the Hodge curvature. Proof B derives it from the same input by clutching propagation from the unconditional genus-1 base case. The two proofs are independent post-input: they invoke disjoint external machinery (GRR/Arakelov vs. Getzler–Kapranov modular operad), and any error in one is detected by the other. The two proofs therefore provide *redundant verification* of the same numerical identity, not *independent derivations* of its structural origin — the latter is furnished uniquely by Lemma 6.5.

Proof A: Grothendieck–Riemann–Roch on the universal curve. This proof identifies the bar-complex obstruction with a characteristic class of the Hodge bundle directly, without induction on the genus. It takes Lemma 6.5 as input.

Step A1: the virtual bundle on moduli. On the uniform-weight lane, the bar complex over a fixed curve Σ_g has a one-dimensional algebraic factor (the scalar channel) tensored with a geometric factor given by the de Rham complex of Σ_g with coefficients in ω_{Σ_g}, because every propagator is the standard weight-1 form $d\log E(z,w)$. As the curve varies over $\overline{\mathcal{M}}_g$, the zero-mode sector of this geometric factor is the direct image $R^0\pi_*\omega_\pi = \mathbb{E}$. The scalar virtual class is therefore

$$[\overline{B}^{(g)}_{\mathrm{scalar}}(\mathcal{A})]^{\mathrm{vir}} \;=\; \kappa(\mathcal{A}) \cdot [\mathbb{E}] \quad \text{in } K^0(\overline{\mathcal{M}}_g) \otimes \mathbb{Q}.$$

Step A2: Mumford's GRR computation. Grothendieck–Riemann–Roch on the universal curve gives

$$\mathrm{ch}(R\pi_*\omega_\pi) \;=\; \pi_*\big(\mathrm{ch}(\omega_\pi) \cdot \mathrm{Td}(T_\pi)\big).$$

The degree-g component of the Chern character of the rank-g bundle $\mathbb{E}$, combined with the Newton identities relating Chern characters to Chern classes, gives

$$c_g(\mathbb{E}) \;=\; \lambda_g \quad \in \mathrm{CH}^g(\overline{\mathcal{M}}_g).$$

This is Mumford's computation [Mumford83].

Step A3: Chern–Weil identification. The Hodge bundle carries the Arakelov metric, whose restriction to each fiber $H^0(\Sigma_g, \Omega^1)$ is the L^2 inner product. The Arakelov–Faltings theorem [Fal84] identifies the curvature of $(\mathbb{E}, h_{\mathrm{Ar}})$ with the Arakelov $(1,1)$-form:

$$\Theta_{\mathbb{E}}\big|_{\mathrm{fiber}} \;=\; \omega_g^{\mathrm{Ar}} \;=\; \frac{i}{2}\sum_{\alpha,\beta=1}^{g} (\mathrm{Im}\,\Omega)^{-1}_{\alpha\beta}\, \omega_\alpha \wedge \overline{\omega}_\beta.$$

The quantum Arnold relations give the fiberwise bar curvature $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g^{\mathrm{Ar}} \cdot \mathrm{id}$. The Arakelov–Faltings theorem identifies ω_g^{Ar} with the curvature of the Hodge bundle. Chern–Weil then yields

$$\mathrm{obs}_g(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \cdot \det\Big(\frac{i}{2\pi}\Theta_{\mathbb{E}}\Big) \;=\; \kappa(\mathcal{A}) \cdot c_g(\mathbb{E}) \;=\; \kappa(\mathcal{A}) \cdot \lambda_g.$$

No clutching argument is used. The only external inputs are Mumford's GRR computation of the Chern classes of $\mathbb{E}$ and the Arakelov–Faltings identification of the Hodge curvature.

Proof B: clutching propagation from genus 1. This proof uses the boundary stratification of $\overline{\mathcal{M}}_g$ and proceeds by induction on g.

Base case. $\mathrm{obs}_1 = \kappa \cdot \lambda_1$ is proved unconditionally (Step 1 above; $H^2(\overline{\mathcal{M}}_{1,1}) \cong \mathbb{Q} \cdot \lambda_1$ is one-dimensional).

Inductive step. Assume $\mathrm{obs}_h = \kappa \cdot \lambda_h$ for all $h < g$. The Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A}) \,\widehat{\otimes}\, \mathfrak{G}^{\mathrm{mod}})$ automatically satisfies the clutching factorization by the Getzler–Kapranov correspondence [GK98, Theorem 5.4]: the MC equation on the Feynman transform encodes exactly the clutching compatibility conditions.

The separating clutching $\xi_{\mathrm{sep}} \colon \overline{\mathcal{M}}_{g_1,1} \times \overline{\mathcal{M}}_{g_2,1} \to \overline{\mathcal{M}}_g$ (with $g_1 + g_2 = g$) gives

$$\xi_{\mathrm{sep}}^*(\mathrm{obs}_g) \;=\; \sum_{g_1+g_2=g} \mathrm{obs}_{g_1} \star \mathrm{obs}_{g_2} \;=\; \sum_{g_1+g_2=g} \kappa \cdot \lambda_{g_1} \star \kappa \cdot \lambda_{g_2} \;=\; \kappa \cdot \xi_{\mathrm{sep}}^*(\lambda_g),$$

where the star product in the convolution algebra contracts along the glued edge via the propagator $\eta^{-1} = 1/\kappa$, absorbing one factor of κ: explicitly, $(\kappa\lambda_{g_1}) \star (\kappa\lambda_{g_2}) = (1/\kappa) \cdot \kappa\lambda_{g_1} \cdot \kappa\lambda_{g_2} = \kappa \cdot \lambda_{g_1}\lambda_{g_2}$. The last equality uses the Whitney sum formula $\xi_{\mathrm{sep}}^*(\lambda_g) = \sum_{i+j=g} \lambda_i \otimes \lambda_j$ for the Hodge bundle.

The non-separating clutching $\xi_{\mathrm{nsep}} \colon \overline{\mathcal{M}}_{g-1,2} \to \overline{\mathcal{M}}_g$ gives

$$\xi_{\mathrm{nsep}}^*(\mathrm{obs}_g) \;=\; \kappa \cdot \xi_{\mathrm{nsep}}^*(\lambda_g),$$

by the inductive hypothesis applied to the genus-$(g-1)$ data at the self-node vertex.

The trace functional (integration over $\overline{\mathcal{M}}_g$) satisfies $\int_{\overline{\mathcal{M}}_g} \mathrm{obs}_g = F_g = \kappa \cdot \hat{A}_g$ (from the bar-complex graph sum on the uniform-weight lane), and $\int_{\overline{\mathcal{M}}_g} \lambda_g = \hat{A}_g$ (the Faber–Pandharipande evaluation). Hence

$$\ell_g(\mathrm{obs}_g) \;=\; \kappa \cdot \ell_g(\lambda_g).$$

It remains to show that the joint map $J_g = (\ell_g, \mathrm{clut}_g) \colon W_g \to \mathbb{C} \oplus B_g$ is injective on the relevant subspace $U_g \ni \mathrm{obs}_g, \lambda_g$. On the uniform-weight lane, U_g is the one-dimensional tautological line $\mathbb{Q} \cdot \lambda_g$ (the scalar channel produces only classes built from the Hodge bundle; no higher Hodge bundles $R^0\pi_*\omega_\pi^{\otimes h}$ with $h \neq 1$ enter). On this one-dimensional space, the trace ℓ_g is already injective (since $\ell_g(\lambda_g) = \hat{A}_g \neq 0$), so J_g is injective a fortiori.

By the tautological-line support criterion, the injectivity of J_g together with the clutching and trace compatibilities gives $\mathrm{obs}_g = \kappa \cdot \lambda_g$.

Agreement of the two proofs. Proof A produces the identification directly from the Hodge bundle structure on the universal curve. Proof B produces it inductively from the boundary stratification. The two proofs have no logical dependency on each other: Proof A uses GRR and Arakelov–Faltings; Proof B uses Getzler–Kapranov and the tautological-line support criterion. Their agreement is a consistency check on the framework. □

Remark 6.7 (Resolution of the clutching-uniqueness gap). The all-genera extension of Theorem D was identified as depending on a "clutching-uniqueness" step: the assertion that the boundary restrictions of obs_g determine obs_g uniquely within the tautological ring. Proof B above establishes this step by showing that the relevant subspace U_g is one-dimensional on the uniform-weight lane (the scalar channel sees only the Hodge bundle), so injectivity of the joint map is automatic. Proof A bypasses the question entirely. The all-genera statement of Theorem D(ii) is therefore unconditional on the uniform-weight lane, via either route.

Example 6.8 (Heisenberg at all genera)**.** $\kappa(\mathcal{H}_k) = k$ (UNIFORM-WEIGHT, single generator of conformal weight 1). The generating function:

$$\sum_{g\geq 1} F_g(\mathcal{H}_k)\cdot \hbar^{2g} \;=\; k\cdot\left(\frac{\hbar/2}{\sin(\hbar/2)} - 1\right) \;=\; \frac{k}{24}\,\hbar^2 + \frac{7k}{5760}\,\hbar^4 + \frac{31k}{967680}\,\hbar^6 + \cdots$$

The chiral partition function $Z_{\mathrm{ch}}(\mathcal{H}_k) = 1/\eta(\tau)$ (where $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$) gives $F_1 = k/24$, consistent with the universal formula.

7 Theorem H: chiral Hochschild cohomology

7.1 Statement

Theorem D identifies a universal genus expansion governed by κ; the question is what constrains the coefficient ring in which κ lives.

Theorem 7.1 (Theorem H: polynomial Hilbert series)**.** *Let $\mathcal{A}$ be a chirally Koszul chiral algebra on a smooth curve X. The chiral Hochschild cohomology is concentrated in degrees $\{0,1,2\}$:*

$$\mathrm{ChirHoch}^n(\mathcal{A}) = 0 \quad \text{for } n \notin \{0,1,2\},$$

and the Hilbert polynomial is

$$P_{\mathcal{A}}(t) \;=\; \dim Z(\mathcal{A}) + \dim \mathrm{ChirHoch}^1(\mathcal{A})\cdot t + \dim Z(\mathcal{A}^!)\cdot t^2. \tag{18}$$

Warning 7.2 (Three Hochschild theories: chiral, topological, categorical). Three inequivalent Hochschild cohomology theories coexist: chiral, topological, and categorical. *Chiral Hochschild* $\mathrm{ChirHoch}^*(\mathcal{A}) = H^*(\mathrm{CoDer}(\overline{B}(\mathcal{A})))$ is the cohomology of coderivations of the bar coalgebra; it lives on the curve X and has amplitude $[0,2]$ on the Koszul locus (Theorem H). *Topological Hochschild* $\mathrm{HH}^*_{\mathrm{top}}$ is factorization homology on S^1; it lives in one dimension higher. *Categorical Hochschild* $\mathrm{HH}^*(\mathrm{Rep}(\mathcal{A}))$ governs the Drinfeld center and the E_2 structure. Conflating any two produces errors.

7.2 Family-specific specializations

Proposition 7.3 (Hilbert series for standard families)**.** (i) Heisenberg $\mathcal{H}_k$: $P(t) = 1 + t + t^2$. *Centre* $= \mathbf{C}$*, outer derivation* $D(J) = \mathbf{1}$*, level deformation in degree* 2.

(ii) Affine Kac–Moody $V_k(\mathfrak{g})$ (generic level): $\mathrm{ChirHoch}^1(V_k(\mathfrak{g})) \cong \mathfrak{g}$*, giving total dimension* $\dim(\mathfrak{g}) + 2$*.* $P(t) = 1 + \dim(\mathfrak{g})\cdot t + t^2$*.*

(iii) Virasoro Vir_c (generic): $P(t) = 1 + t^2$. $\mathrm{ChirHoch}^1 = 0$ *(no outer chiral derivations).*

(iv) Principal $\mathcal{W}_N$ (generic): $P(t) = 1 + t^2$.

(v) Critical level $k = -h^\vee$: *Outside the regime of Theorem H. The Feigin–Frenkel center makes the cohomology unbounded.*

7.3 Proof architecture

The descent lemma reduces Theorem H to the Verdier intertwining of Theorem A: the concentration of $\mathrm{ChirHoch}^*(\mathcal{A})$ in degrees $\{0, 1, 2\}$ follows from the Koszul resolution having length 2 (the bar coalgebra has generators in a single bar degree on the Koszul locus), and the identification of the Hilbert polynomial with the dimensions of the center and its dual follows from the Verdier intertwining pairing degree-0 with degree-2.

The configuration-space collapse uses Fulton–MacPherson formality: the spectral sequence from the FM filtration on the coderivation complex collapses, and the E_2 page identifies with the cohomology of the classical coderivation complex of the associated graded.

8 The forced chain A ⇒ B ⇒ C ⇒ D ⇒ H

The five theorems are not five unrelated assertions about the bar complex. They form a chain: each theorem solves a problem created by its predecessor, and each requires exactly one new geometric input beyond what the previous theorem provides.

8.1 The dependency DAG

The logical dependencies are strictly linear:

$$\boxed{\text{A}} \xrightarrow{\text{dual object}} \boxed{\text{B}} \xrightarrow{\text{coexistence}} \boxed{\text{C}} \xrightarrow{\text{polarization}} \boxed{\text{D}} \xrightarrow{\text{coefficient ring}} \boxed{\text{H}}$$

No arrow can be reversed. Each node depends on all preceding nodes. The chain is *forced*: removing any one theorem leaves the subsequent ones either undefined or vacuous.

8.2 A creates the arena

Proposition 8.1 (A is necessary for B–H)**.** *Without Theorem A, the objects of Theorems B–H do not exist.*

Proof. Theorem B asserts that the bar-cobar counit $\varepsilon\colon \Omega(\overline{B}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism. The cobar functor Ω is defined on conilpotent coalgebras in the factorization category $\mathrm{Fact}(X)$; its existence and the adjunction with $\overline{B}$ are the content of Theorem A(i). Without the adjunction, the counit ε is undefined.

Theorem C decomposes $H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}}) \cong Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!)$. The summand $Q_g(\mathcal{A}^!)$ involves the Koszul dual $\mathcal{A}^!_\infty$, whose existence is Theorem A(ii): the Verdier intertwining $\mathbb{D}_{\mathrm{Ran}}\,\overline{B}(\mathcal{A}) \simeq \mathcal{A}^!_\infty$. Without this intertwining, the dual object is unavailable.

Theorems D and H depend on C and B respectively; their dependence on A is inherited. □

The geometric input that Theorem A requires, beyond the bar complex itself, is the Fulton–MacPherson compactification $\overline{\mathrm{FM}}_n(X)$ and its boundary divisor structure: the twisting morphism τ is mediated by integration kernels on $\overline{\mathrm{FM}}_n(X)$, and the Verdier intertwining uses the Poincaré–Verdier duality on $\mathrm{Ran}(X)$.

8.3 B establishes faithfulness

Proposition 8.2 (B is necessary for C and strict-dual identification)**.** *Without Theorem B, the Lagrangian decomposition of Theorem C is vacuous and the relation between the homotopy dual and the strict Koszul dual is ungrounded.*

Proof. Theorem A constructs the homotopy Koszul dual $\mathcal{A}^!_\infty$, but the construction produces a factorization algebra that a priori differs from $\mathcal{A}^!$. Theorem B proves that bar-cobar inversion is faithful: $\overline{B}$ is injective on quasi-isomorphism classes. This identifies $\mathcal{A}$ and $\mathcal{A}^!$ as the two canonical algebras recoverable from the bar coalgebra, and establishes that Verdier duality on $\overline{B}(\mathcal{A})$ produces the *correct* dual.

Without this identification, the decomposition $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!)$ in Theorem C involves an object $\mathcal{A}^!_\infty$ whose relation to the strict dual $\mathcal{A}^!$ is uncontrolled: the summands may not be complementary. Theorem D is bar-intrinsic and does not use this identification to produce the scalar genus tower, but the strict-dual interpretation of complementarity and the downstream duality package do require it. □

The geometric input is the *PBW filtration*: the spectral sequence from conformal-weight filtration collapses at E_2, reducing the problem to classical Koszul duality of the associated graded. This is the only step that involves the conformal weight grading; Theorem A uses only the chiral bracket and $\mathcal{D}_X$-module structure.

8.4 C provides the polarization

Proposition 8.3 (C is necessary for the ambient polarization package)**.** *Without Theorem C, the genus tower has no canonical factorization into $\mathcal{A}$ and $\mathcal{A}^!$ contributions, and the complementarity interpretation of the scalar genus tower is unavailable.*

Proof. At genus $g \geq 1$, the bar differential satisfies $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$, and the Maurer–Cartan element $\Theta_{\mathcal{A}}$ encodes all genus corrections. But the ambient complex $\mathbf{C}_g(\mathcal{A}) = R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ receives contributions from both $\mathcal{A}$ and its dual. Without the Lagrangian decomposition, these contributions are entangled: one loses the canonical ambient polarization into $\mathcal{A}$ and $\mathcal{A}^!$ sectors.

Theorem C provides the polarization: the Verdier involution decomposes $\mathbf{C}_g$ into eigenspaces, and the scalar projection of this decomposition yields the complementarity constraint $\kappa + \kappa^! = \varrho_{\mathcal{A}} \cdot K$ (family-specific $\varrho_{\mathcal{A}}$: 0 for Kac–Moody and free fields, 1/2 for Virasoro, $H_N - 1$ for principal $\mathcal{W}_N$, 1/6 for Bershadsky–Polyakov). Theorem D still extracts the universal scalar shadow bar-intrinsically from the genus-1 curvature together with clutching/GRR propagation; Theorem C explains how that scalar sits inside the ambient complex and how duality constrains it. □

The geometric input is the *Verdier involution*: the existence of a functorial duality on sheaves on $\overline{\mathcal{M}}_g$ that exchanges the bar complex of $\mathcal{A}$ with that of $\mathcal{A}^!$. This requires the moduli space $\overline{\mathcal{M}}_g$ as a smooth Deligne–Mumford stack with Poincaré duality.

8.5 D identifies the scalar shadow

Proposition 8.4 (D is necessary for H)**.** *Without Theorem D, the coefficient ring of the genus tower is unidentified, and Theorem H cannot distinguish polynomial behaviour from more complex coefficient growth.*

Proof. The genus expansion $\sum_g F_g(\mathcal{A}) \cdot \hbar^{2g} = \kappa \cdot (\hat{A}(i\hbar) - 1)$ shows that all genus invariants are determined by a single number κ. The Maurer–Cartan element $\Theta_{\mathcal{A}}$ lives in the convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$, whose coefficient ring is the chiral Hochschild cohomology $\mathrm{ChirHoch}^*(\mathcal{A})$. Theorem D demonstrates that the genus tower is not an independent sequence of invariants but a single series governed by κ; this forces the question: what constrains κ itself?

The answer is Theorem H. The modular characteristic κ lives in degree 2 of $\mathrm{ChirHoch}^*(\mathcal{A})$. The deformations of κ are parametrized by $\mathrm{ChirHoch}^2(\mathcal{A})$; the obstructions are in $\mathrm{ChirHoch}^3(\mathcal{A})$. Theorem H proves $\mathrm{ChirHoch}^3 = 0$: the deformation theory is unobstructed, and the coefficient ring is polynomial. Without Theorem D's identification of κ as the universal scalar, one would not know that the coefficient ring has a single generator in degree 2; one would only know that genus invariants exist. □

The geometric input is *Faber–Pandharipande integration*: the evaluation of λ_g on $\overline{\mathcal{M}}_g$ produces the numerical coefficients $\hat{A}_g$, and the factorization through stable graphs uses the clutching morphisms on the boundary of $\overline{\mathcal{M}}_g$.

8.6 The non-circular routing

The chain A ⇒ B ⇒ C ⇒ D ⇒ H might appear circular: Theorem A uses the bar complex, whose higher-genus structure involves κ (which is computed in Theorem D). The resolution is that the chain has a non-circular anchor.

Remark 8.5 (Primitive non-circular anchor)*.* The bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ and the Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$ are defined intrinsically, without reference to any theorem. The MC equation $d_0\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ is a tautology of the bar differential $D^2_{\mathcal{A}} = 0$.

Theorem A uses only genus-0 data: the twisting morphism and the Verdier functor on $\mathrm{Ran}(X)$. Theorem B uses the PBW filtration at genus 0. Theorem C uses the Verdier involution on $\overline{\mathcal{M}}_g$ applied to the objects constructed in A and identified in B. Theorem D uses the bar-intrinsic scalar trace of $\Theta_{\mathcal{A}}$, the genus-1 curvature normalization, and the clutching/GRR propagation to identify the scalar shadow κ, which was already present in the definition of $\Theta_{\mathcal{A}}$ but not yet identified as universal. Theorem C gives a complementary ambient interpretation of the same scalar, but is not the load-bearing input to Theorem D. Theorem H uses the concentration result from Koszul duality (Theorem B) together with the scalar package of Theorem D to bound the coderivation complex.

The routing is: bar-intrinsic MC element (definition) → adjunction/intertwining (A), then two downstream branches. One branch is inversion (B) → Verdier polarization (C). The other is scalar trace + genus-1 curvature + clutching/GRR → scalar extraction (D). Theorem H reads the coefficient ring against these outputs. No step requires a result proved later.

8.7 Independence of converses

The converses of the five implications fail. This is not a deficiency; it reflects the fact that each theorem addresses a strictly different aspect of the bar complex.

1. *H does not imply D.* A polynomial Hilbert series $P(t) = 1 + t^2$ (Virasoro) does not determine $\kappa = c/2$; it constrains $\dim \mathrm{ChirHoch}^2 = 1$ (the space where κ lives has dimension 1) but not the value.

2. *D does not imply C.* The formula $\mathrm{obs}_g = \kappa \cdot \lambda_g$ could hold without a Lagrangian decomposition: one could imagine a non-decomposable ambient complex whose scalar projection still factors through κ. Theorem C is stronger; it provides a complementary ambient interpretation of the factorization, but not the load-bearing proof of Theorem D.

3. *C does not imply B.* The Lagrangian decomposition $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!)$ is meaningful even if bar-cobar inversion fails; one could have a Lagrangian complement indexed by a coalgebra that does not reconstruct $\mathcal{A}$. Inversion is a separate condition.

4. *B does not imply A.* One could define bar-cobar inversion without the adjunction formalism: the counit ε could be a quasi-isomorphism without being part of an adjoint pair. The adjunction gives functoriality; inversion alone does not.

8.8 Summary: one new input per step

Theorem	**Output**	**New input**	**Problem solved**
A	dual object $\mathcal{A}^!_\infty$	FM boundary	no dual without adjunction
B	inversion $\Omega(\overline{B}(\mathcal{A})) \simeq \mathcal{A}$	PBW filtration	bar could lose boundary algebra
C	$Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!)$	Verdier involution on $\overline{\mathcal{M}}_g$	no canonical polarization
D	$\mathrm{obs}_g = \kappa \cdot \lambda_g$	Faber–Pandharipande	no distinguished scalar shadow
H	$P(t) = 1 + \dim(\mathfrak{g}) \cdot t + t^2$	coderivation complex	coefficient ring uncontrolled

Each row adds exactly one geometric tool to the preceding rows. The five-step chain is the minimum path from the bare bar complex to the complete structural description of the modular Koszul dual.

9 The Heisenberg: complete computation

The Heisenberg algebra $\mathcal{H}_k$ is the simplest chiral algebra that is not a Lie algebra. It has one generator, one double pole, and no simple pole. Every clause of every theorem is computable in closed form. This section carries out the computation: the five theorems verified in a single place, with no step deferred.

9.1 The algebra

The Heisenberg chiral algebra $\mathcal{H}_k$ at level $k \in \mathbf{C}$ is generated by a single current J of conformal weight 1, with the operator product expansion

$$J(z)\, J(w) \;\sim\; \frac{k}{(z-w)^2}\,. \tag{19}$$

One generator, one OPE pole of order 2, no simple pole. The zeroth product $J_{(0)}J = 0$ (no Lie bracket: the algebra is abelian). The first product $J_{(1)}J = k$ (the level is the inner product). The augmentation ideal is $\overline{\mathcal{H}}_k = \mathbf{C} \cdot J$, one-dimensional.

The classical R-matrix is scalar:

$$r^{\mathcal{H}}(z) \;=\; \frac{k}{z}\,, \tag{20}$$

obtained from the OPE double pole by $d\log$ absorption: $\mathrm{pole}_r = \mathrm{pole}_{\mathrm{OPE}} - 1 = 2 - 1 = 1$. The modular characteristic is $\kappa(\mathcal{H}_k) = \mathrm{av}(r^{\mathcal{H}}(z)) = k$. Since $r(z)$ is already scalar, the averaging map is the identity: no matrix structure to erase, no Casimir tensor to contract. The Heisenberg is the unique family where the ordered and symmetric bars carry exactly the same information.

9.2 Theorem A for $\mathcal{H}_k$: bar-cobar adjunction

The bar complex is the cofree coalgebra on the desuspended augmentation ideal:

$$\overline{B}(\mathcal{H}_k) \;=\; T^c(s^{-1}\mathbf{C}\cdot J) \;=\; \bigoplus_{n\geq 1}(s^{-1}J)^{\otimes n}, \qquad |s^{-1}J| = 1-1 = 0. \tag{21}$$

The bar differential at degree 2 is the complete computation:

$$d_{\overline{B}}[s^{-1}J\,|\,s^{-1}J] \;=\; \mathrm{Res}_{z_1=z_2}\left[\frac{k}{(z_1-z_2)^2}\cdot d\log(z_1-z_2)\right] \;=\; k. \tag{22}$$

At degree $n \geq 3$, the bar differential involves iterated residues that reduce to degree $n-1$ by the Arnold relation. The bar cohomology concentrates in bar degree 1:

$$H^*(\overline{B}(\mathcal{H}_k)) \;=\; \mathbf{C}\cdot s^{-1}J \qquad (\text{degree } -1).$$

The Koszul dual coalgebra is $\mathcal{H}_k^{\text{i}} = \mathbf{C}\cdot s^{-1}J$ (one-dimensional, in degree -1). The strict Koszul dual is the curved symmetric chiral algebra

$$\mathcal{H}_k^! \;=\; \mathrm{Sym}^{\mathrm{ch}}(V^*), \qquad m_0 = -k\omega, \tag{23}$$

with curvature $m_0 = -k\omega$ encoding the dual level. The Verdier intertwining of Theorem A(ii) recovers $\mathcal{H}_k^!$ at the cohomological level: $\mathbb{D}_{\mathrm{Ran}}\,\overline{B}(\mathcal{H}_k) \simeq (\mathcal{H}_k^!)_\infty$.

9.3 Theorem B for $\mathcal{H}_k$: bar-cobar inversion

The cobar of the bar recovers the original algebra:

$$\Omega(\overline{B}(\mathcal{H}_k)) \;\xrightarrow{\sim}\; \mathcal{H}_k. \tag{24}$$

The PBW spectral sequence collapses at E_2 because the bar cohomology is already concentrated in degree 1: the associated graded is quadratic Koszul in the classical sense (the relation space is one-dimensional, generated by $J\otimes J - k$).

Bar-cobar inversion for $\mathcal{H}_k$ is *unconditional at all genera.* At genus $g \geq 1$, the fiberwise bar differential satisfies $d_{\mathrm{fib}}^2 = k\cdot\omega_g$ with scalar curvature. The Ext vanishing condition of Theorem 4.1(ii) is automatic: the OPE has a single double pole and no simple pole, so there are no Lie brackets to obstruct and no higher-order collisions to produce non-trivial Ext groups. The absence of a simple pole ($J_{(0)}J = 0$) is the algebraic reason that $\mathcal{H}_k$ is the simplest non-trivial test case for every theorem in the programme.

9.4 Theorem C for $\mathcal{H}_k$: complementarity

The Koszul dual has modular characteristic $\kappa(\mathcal{H}_k^!) = -k$: the curvature $m_0 = -k\omega$ encodes the opposite level. The complementarity sum is

$$\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) \;=\; k + (-k) \;=\; 0 \;=\; K(\mathcal{H}_k). \tag{25}$$

The Koszul complementarity conductor $K = 0$: the Heisenberg is in the class of algebras where duality reverses the sign of κ.

The Lagrangian decomposition of Theorem 5.1 is visible by inspection. The center local system $\mathcal{Z}_{\mathcal{H}_k}$ on $\overline{\mathcal{M}}_g$ is the constant sheaf (the center of $\mathcal{H}_k$ is $\mathbf{C}$, with no monodromy). The ambient complex $\mathbf{C}_g(\mathcal{H}_k) = R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{H}_k}) = H^*(\overline{\mathcal{M}}_g, \mathbf{C})$ decomposes into the Hodge eigenspaces of the Verdier involution, and the two Lagrangian summands $Q_g(\mathcal{H}_k)$ and $Q_g(\mathcal{H}_k^!)$ are exchanged by $\kappa \mapsto -\kappa$.

9.5 Theorem D for $\mathcal{H}_k$: the genus expansion

The Heisenberg is a uniform-weight algebra (single generator of conformal weight 1), so the genus expansion takes the unconditional form

$$\mathrm{obs}_g(\mathcal{H}_k) \;=\; k \cdot \lambda_g \qquad (g \geq 1) \quad \text{(UNIFORM-WEIGHT)}. \tag{26}$$

No cross-channel correction: $\delta F_g^{\mathrm{cross}} = 0$ because there is only one conformal weight.

The genus-1 free energy:

$$F_1(\mathcal{H}_k) \;=\; \frac{k}{24}\,. \tag{27}$$

The full generating function is

$$\sum_{g \geq 1} F_g(\mathcal{H}_k) \cdot \hbar^{2g} \;=\; k \cdot \left(\frac{\hbar/2}{\sin(\hbar/2)} - 1 \right) \;=\; \frac{k}{24}\,\hbar^2 + \frac{7k}{5760}\,\hbar^4 + \frac{31k}{967680}\,\hbar^6 + \cdots \tag{28}$$

with coefficients $F_g(\mathcal{H}_k) = k \cdot \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$, where B_{2g} is the $2g$-th Bernoulli number.

Remark 9.1 (The chiral partition function). The genus-1 chiral partition function of $\mathcal{H}_k$ is

$$Z_{\mathrm{ch}}(\mathcal{H}_k) \;=\; \frac{1}{\eta(\tau)}\,, \qquad \eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n),$$

where $q = e^{2\pi i \tau}$. The leading exponent $q^{-1/24}$ encodes $F_1 = k/24$ (at $k = 1$). The product form is the Euler function; its reciprocal counts states at each energy level.

Remark 9.2 (R-matrix and averaging). The R-matrix $r^{\mathcal{H}}(z) = k/z$ is already scalar, so $\mathrm{av}(r^{\mathcal{H}}(z)) = k = \kappa(\mathcal{H}_k)$ with no information lost. For every other family, the averaging map av compresses a matrix-valued function of z into a single number. The Heisenberg is the unique base case where the compression is lossless. For non-abelian affine Kac–Moody, the averaging discards the Casimir tensor Ω and the z-dependence; for the Virasoro algebra, it discards the stress tensor T and the cubic pole structure. The ordered bar sees the full R-matrix; the symmetric bar sees only κ.

9.6 Theorem H for $\mathcal{H}_k$: chiral Hochschild cohomology

The chiral Hochschild cohomology of $\mathcal{H}_k$ is three-dimensional:

$$\mathrm{ChirHoch}^n(\mathcal{H}_k) = \begin{cases} \mathbf{C} & n = 0, \\ \mathbf{C} & n = 1, \\ \mathbf{C} & n = 2, \\ 0 & n \geq 3. \end{cases} \tag{29}$$

The Hilbert polynomial is $P_{\mathcal{H}_k}(t) = 1 + t + t^2$.

The three generators have transparent algebraic meaning:

1. *Degree* 0*: the center.* $\mathrm{ChirHoch}^0(\mathcal{H}_k) = Z(\mathcal{H}_k) = \mathbf{C}$. The center consists of scalars; J has no zeroth product ($J_{(0)} = 0$), so J itself is central, but J is not a scalar; the degree-0 chiral Hochschild class is the identity.

2. *Degree* 1*: the outer derivation.* $\mathrm{ChirHoch}^1(\mathcal{H}_k) = \mathbf{C}$, generated by the derivation $D\colon J \mapsto \mathbf{1}$ that maps the current to the vacuum. This derivation is outer: it is not of the form $a \mapsto a_{(0)}b$ for any $b \in \mathcal{H}_k$, because $J_{(0)} = 0$ (no simple pole in the OPE). The existence of an outer derivation is the Heisenberg's distinguishing feature relative to the Virasoro algebra, where $\mathrm{ChirHoch}^1 = 0$.

3. *Degree* 2*: the level deformation.* $\mathrm{ChirHoch}^2(\mathcal{H}_k) = \mathbf{C}$, generated by the infinitesimal deformation of the OPE coefficient $k \mapsto k + \epsilon$. This is the dual of the center: the Verdier pairing identifies $\mathrm{ChirHoch}^2(\mathcal{H}_k) \cong Z(\mathcal{H}_k^!)^\vee = \mathbf{C}$, consistent with the general formula (18).

9.7 The shadow tower and the derived center

The shadow tower $\{S_r(\mathcal{H}_k)\}_{r\geq 2}$ terminates at $r = 2$:

$$S_2 = \kappa = k, \qquad S_r = 0 \text{ for all } r \geq 3.$$

The critical discriminant $\Delta = 8\kappa S_4 = 0$: the Heisenberg is class G (Gaussian), the simplest archetype. The shadow tower carries no information beyond κ itself.

The derived chiral center is $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{H}_k) = \mathrm{ChirHoch}^*(\mathcal{H}_k) = \mathbf{C}\oplus\mathbf{C}[1]\oplus\mathbf{C}[2]$, three-dimensional ($k \neq 0$). This is strictly larger than the naive center $Z(\mathcal{H}_k) = \mathbf{C}$: the Ext^1 and Ext^2 contributions are invisible to the commutant.

Remark 9.3 (Abelian Chern–Simons). The Heisenberg algebra $\mathcal{H}_k$ is the boundary algebra of abelian Chern–Simons theory at level k: the 3-dimensional bulk is $U(1)_k$ Chern–Simons on a 3-manifold M^3 with $\partial M^3 = \Sigma$. The modular characteristic $\kappa = k$ is the Chern–Simons level; the genus expansion $F_g = k \cdot \hat{A}_g$ is the perturbative expansion of the Chern–Simons partition function. The complementarity sum $K = 0$ reflects the topological nature of the theory: bulk and boundary carry opposite orientations.

9.8 Summary: the Heisenberg as base computation

Theorem	Statement for $\mathcal{H}_k$	Key simplification
A	$\overline{B}(\mathcal{H}_k) = T^c(s^{-1}\mathbf{C} \cdot J)$, $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$	one generator, degree-1 concentration
B	$\Omega(\overline{B}(\mathcal{H}_k)) \simeq \mathcal{H}_k$, all genera	no Lie bracket, Ext automatic
C	$\kappa + \kappa' = 0$	Verdier reverses sign
D	$F_g = k \cdot \hat{A}_g$ (UNIFORM-WEIGHT)	single weight, no cross-channel
H	$P(t) = 1 + t + t^2$	outer derivation from abelian OPE

Every other family is a deformation of this computation. Affine Kac–Moody adds a simple pole (Lie bracket); the Virasoro algebra adds a quartic pole (stress tensor) and the cubic R-matrix structure; the $\mathcal{W}$-algebras add higher-spin generators and multi-channel interactions. The five theorems persist, but their proofs become harder: bar cohomology is still concentrated in degree 1 (Koszul), but the spectral sequence argument requires PBW-type tools; the complementarity sum is no longer zero; the genus expansion acquires cross-channel corrections for multi-weight algebras; the chiral Hochschild loses its degree-1 component. The Heisenberg computation is the base case that makes the pattern visible.

10 The standard landscape

The five theorems hold for every chirally Koszul chiral algebra. The *standard landscape* is the collection of families for which every invariant has been computed: the modular characteristic κ, the Koszul complementarity conductor K, the critical discriminant Δ, the complexity class, the shadow depth $r_{\max}$, and the classical R-matrix. This section records the census.

10.1 The census table

In the table: $H_N = \sum_{j=1}^{N} 1/j$ is the N-th harmonic number; $K = \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$ is the Koszul complementarity conductor; $\Delta = 8\kappa S_4$ is the critical discriminant; $r_{\max}$ is the shadow depth (the highest degree at which $S_r \neq 0$, or ∞ if the tower does not terminate).

10.2 The general affine Kac–Moody formula

For a simple Lie algebra $\mathfrak{g}$ of dimension $d = \dim \mathfrak{g}$ and dual Coxeter number $h^\vee$, the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ at level k has

$$\kappa(V_k(\mathfrak{g})) \;=\; \frac{d \cdot (k + h^\vee)}{2h^\vee}\,. \tag{30}$$

The formula is derived from the trace-form classical R-matrix $r(z) = k\,\Omega/z$:

$$\mathrm{av}(r(z)) \;=\; \frac{k \cdot d}{2h^\vee} \;=\; \kappa - \frac{d}{2}\,,$$

where the shift $d/2$ is the Sugawara contribution from the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$. The full modular characteristic is $\kappa = \mathrm{av}(r(z)) + d/2 = d(k + h^\vee)/(2h^\vee)$.

Two boundary checks:

1. At $k = 0$ (abelian limit): $\kappa(V_0(\mathfrak{g})) = d/2 \neq 0$. The R-matrix $r(z) = 0$ vanishes, but the Sugawara shift persists.
2. At $k = -h^\vee$ (critical level): $\kappa(V_{-h^\vee}(\mathfrak{g})) = 0$. All monodromy becomes trivial; the Feigin–Frenkel center appears; Koszulness fails.

The Koszul duality acts by $k \mapsto -k-2h^\vee$ (the Feigin–Frenkel involution), giving $\kappa(V_k(\mathfrak{g})^!) = -d(k+h^\vee)/(2h^\vee) = -\kappa$, so $K = 0$ for all affine Kac–Moody algebras. The complementarity sum $c + c'$ is $2d$, independent of k.

Example 10.1 (Specific Lie algebras).

$\mathfrak{g}$	d	$h^\vee$	κ	$c+c'$	$\kappa\|_{k=0}$
$\mathfrak{sl}_2$	3	2	$3(k+2)/4$	6	3/2
$\mathfrak{sl}_3$	8	3	$4(k+3)/3$	16	4
$\mathfrak{so}_8$	28	6	$7(k+6)/3$	56	14
G_2	14	4	$7(k+4)/4$	28	7
F_4	52	9	$26(k+9)/9$	104	26
E_6	78	12	$13(k+12)/4$	156	39
E_7	133	18	$133(k+18)/36$	266	133/2
E_8	248	30	$62(k+30)/15$	496	124

10.3 The R-matrix census

The classical R-matrix $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is the binary collision residue. Its pole structure is determined by the OPE pole order via $d\log$ absorption:

$$\mathrm{pole}_r \;=\; \mathrm{pole}_{\mathrm{OPE}} - 1. \tag{31}$$

Family	**OPE pole**	**r-matrix pole**	$r(z)$
$\mathcal{H}_k$	2	1	k/z (scalar)
$V_k(\mathfrak{g})$	2	1	$k\,\Omega/z$ (matrix, trace-form)
Vir_c	4	3	$(c/2)/z^3 + 2T/z$ (cubic + simple)
$\mathcal{W}_N$	$2N$	$2N-1$	matrix-valued, $N-1$ channels
$\beta\gamma$	1	0	constant
bc	1	0	Clifford-valued

The Virasoro R-matrix has a cubic pole and a simple pole. The cubic pole encodes the central charge; the simple pole encodes the stress tensor. It is *not* quartic: the quartic pole of the $T(z)\,T(w)$ OPE is absorbed by $d\log$.

10.4 The four complexity classes

The shadow obstruction tower $\{S_r\}_{r\geq 2}$ partitions the landscape into four classes. The partition is determined by two invariants: S_4 (the quartic shadow coefficient) and κ.

Definition 10.2 (Complexity class)**.** Let $\Delta = 8\kappa S_4$ be the critical discriminant. The four classes are:

(a) **Class G** (Gaussian): $S_r = 0$ for all $r \geq 3$. Shadow depth $r_{\max} = 2$. The shadow tower carries only κ; no nonlinear corrections. $\Delta = 0$ because $S_4 = 0$.

(b) **Class L** (Lie): $S_3 \neq 0$, $S_4 = 0$. Shadow depth $r_{\max} = 3$. A single Massey product appears at degree 3 (the Lie bracket), then the tower terminates. $\Delta = 0$ because $S_4 = 0$.

(c) **Class C** (Contact): $S_3 = 0$, $S_4 \neq 0$, but $\kappa = 0$ or special cancellation gives $\Delta = 0$. Shadow depth $r_{\max} = 4$. The $\beta\gamma$ system is the witness: $r_{\max} = 4$ but $S_r = 0$ for $r \geq 5$.

(d) **Class M** (Mixed): $\Delta \neq 0$. Shadow depth $r_{\max} = \infty$. The tower does not terminate; every S_r is nonzero. The Virasoro algebra and all principal $\mathcal{W}_N$-algebras are class M.

The partition is *forced*: the critical discriminant Δ governs a dichotomy. If $\Delta = 0$ (i.e. $\kappa = 0$ or $S_4 = 0$), the MC equation at degree 4 is automatically satisfied, and the tower can terminate. If $\Delta \neq 0$, the degree-4 MC equation produces a non-trivial S_5, which feeds degree 5 to produce S_6, and so on: the tower is infinite. There is no class with shadow depth 3 or any finite value ≥ 3 other than 4: the algebraic depth $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ has a gap.

Class	S_3	S_4	Δ	**Families**
G	0	0	0	Heisenberg, lattice, free fermion, bc
L	$\neq 0$	0	0	Affine Kac–Moody (all $\mathfrak{g}$, generic k)
C	0	$\neq 0$	0	$\beta\gamma$
M	$\neq 0$	$\neq 0$	$\neq 0$	Virasoro, $\mathcal{W}_N$, Bershadsky–Polyakov

10.5 Free-field central charges

The free-field families provide essential boundary data.

Fermionic bc ghosts (conformal weights λ and $1 - \lambda$):

$$c_{bc}(\lambda) \;=\; 1 - 3(2\lambda - 1)^2. \tag{32}$$

At $\lambda = 1/2$: $c_{bc} = 1$ (a single Dirac fermion). At $\lambda = 2$: $c_{bc} = -26$ (the reparametrization ghosts of the bosonic string).

Bosonic $\beta\gamma$ system (conformal weights λ and $1 - \lambda$):

$$c_{\beta\gamma}(\lambda) \;=\; 2(6\lambda^2 - 6\lambda + 1). \tag{33}$$

At $\lambda = 1/2$: $c_{\beta\gamma} = -1$ (the symplectic boson). At $\lambda = 2$: $c_{\beta\gamma} = 26$.

The complementarity relation

$$c_{\beta\gamma}(\lambda) + c_{bc}(\lambda) \;=\; 0 \qquad \text{for all } \lambda \tag{34}$$

is the ghost cancellation: bosonic and fermionic partners contribute opposite central charges. At $\lambda = 2$, this is the string-theoretic statement $26 + (-26) = 0$.

10.6 The Virasoro self-dual point

Under Koszul duality, the Virasoro algebra satisfies $c \mapsto 26 - c$. The modular characteristic:

$$\kappa(\mathrm{Vir}_c) = c/2, \qquad \kappa(\mathrm{Vir}_{26-c}) = (26-c)/2. \tag{35}$$

The complementarity sum is $K = c/2 + (26-c)/2 = 13$, constant and independent of c. The self-dual point is

$$c = 13\colon \quad \kappa = \kappa' = 13/2. \tag{36}$$

This is the unique point where the Lagrangian decomposition of Theorem 5.1 is symmetric: the two summands $Q_g(\mathrm{Vir}_{13})$ and $Q_g(\mathrm{Vir}_{13}^!)$ are isomorphic.

Warning 10.3 (The self-dual point is $c = 13$, not $c = 26$). The complementarity sum $c + c' = 26$ involves both central charges; the self-dual *point* is $c = c' = 13$, the fixed point of $c \mapsto 26 - c$. Confusing the sum with the fixed point is a persistent error.

10.7 The $\mathcal{W}_N$ family

The principal $\mathcal{W}$-algebra $\mathcal{W}_N$ is the Drinfeld–Sokolov reduction of $\widehat{\mathfrak{sl}}_N$. It has $N-1$ strong generators of conformal weights $2, 3, \ldots, N$. The modular characteristic:

$$\kappa(\mathcal{W}_N) \;=\; c \cdot (H_N - 1), \qquad H_N = \sum_{j=1}^{N} \frac{1}{j}\,. \tag{37}$$

At $N = 2$: $H_2 - 1 = 3/2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2 = \kappa(\mathrm{Vir}_c)$, consistent with $\mathcal{W}_2 = \mathrm{Vir}$. At $N = 3$: $H_3 - 1 = 1 + 1/2 + 1/3 - 1 = 5/6$, so $\kappa(\mathcal{W}_3) = 5c/6$.

All $\mathcal{W}_N$-algebras with $N \geq 2$ are class M: the shadow tower is infinite and the critical discriminant $\Delta \neq 0$. As multi-weight algebras (generators at weights $2, \ldots, N$), at genus $g \geq 2$ the full free energy acquires a cross-channel correction $\delta F_g^{\mathrm{cross}}$ from mixed conformal-weight interactions. The scalar formula $\mathrm{obs}_g = \kappa \cdot \lambda_g$ is unconditional at genus 1; at higher genera the diagonal term persists but $\delta F_g^{\mathrm{cross}} \neq 0$ (ALL-WEIGHT).

10.8 The Bershadsky–Polyakov algebra

The Bershadsky–Polyakov algebra $\mathrm{BP}_k = \mathcal{W}_3^{(2)}(k)$ is the subregular DS reduction of $\widehat{\mathfrak{sl}}_3$. It has four generators: J (weight 1), $G^{\pm}$ (weight 3/2), and T (weight 2). The central charge:

$$c(\mathrm{BP}_k) = 2 - \frac{24(k+1)^2}{k+3}\,.$$

The modular characteristic $\kappa(\mathrm{BP}_k) = c/6$, and the Koszul complementarity conductor:

$$K(\mathrm{BP}) \;=\; \kappa(\mathrm{BP}_k) + \kappa(\mathrm{BP}_{-k-6}) \;=\; \frac{98}{3}\,. \tag{38}$$

The self-dual level $k = -3$ coincides with the critical level $-h^\vee(\mathfrak{sl}_3)$, where $c(\mathrm{BP}_k)$ has a simple pole; the value $\kappa(\mathrm{BP}_{-3}) = 49/3$ quoted here is the *principal-value symmetric limit* $\lim_{\varepsilon\to 0} \frac{1}{2}[\kappa(\mathrm{BP}_{-3+\varepsilon}) + \kappa(\mathrm{BP}_{-3-\varepsilon})]$ across the pole, not a regular function value. The full Koszul conductor is a polynomial identity in $\mathbb{Q}(k)$: $K_{\mathrm{BP}}(k) := c(k) + c(-k-6) \equiv 196$ as an element of $\mathbb{Q}(k)$; equivalently $c(k) - 98$ is an odd function of $(k+3)$. See Volume I standalone `bp_self_duality.tex`, Theorem `thm:bp-koszul-conductor-polynomial`, for the full statement.

10.9 The Hilbert polynomial census

Proposition 10.4 (Hilbert polynomials across the landscape)**.** *The chiral Hochschild Hilbert polynomial for the standard families:*

Family	$P(t)$	$\dim \mathrm{ChirHoch}^1$
$\mathcal{H}_k$ *(Heisenberg)*	$1+t+t^2$	1
Lattice V_Λ *(*rank $= r$*)*	$1+r\,t+t^2$	r
$V_k(\mathfrak{g})$ *(affine Kac–Moody)*	$1+\dim(\mathfrak{g})\cdot t+t^2$	$\dim(\mathfrak{g})$
Vir_c *(generic)*	$1+t^2$	0
$\mathcal{W}_N$ *(generic)*	$1+t^2$	0
BP_k *(generic)*	$1+t+t^2$	1
$\beta\gamma$	$1+t+t^2$	1
bc	$1+t^2$	0

The degree-1 component $\mathrm{ChirHoch}^1(\mathcal{A})$ is the space of chiral derivations modulo inner derivations. Its dimension distinguishes families; the computation rests on the following Garland–Lepowsky-type identification.

Lemma 10.5 (Chiral H^1 for freely-generated VOAs at generic level)**.** *Let $\mathcal{A}$ be a freely strongly-generated chiral algebra with generating space $V = \bigoplus_i V_{h_i}$ at conformal weights $h_i \geq 1$, at a generic (non-critical, non-resonant) level. Then*

$$\mathrm{ChirHoch}^1(\mathcal{A}) \;\cong\; V_1$$

*is the space of weight-*1 *strong generators. Equivalently,* $\dim \mathrm{ChirHoch}^1(\mathcal{A})$ *equals the number of current-type (spin-*1*) strong generators.*

Proof sketch. Chiral derivations $D\colon \mathcal{A} \to \mathcal{A}$ of weight 0 are determined by their values on strong generators. At generic level, the resolution $\overline{B}(\mathcal{A})$ is concentrated in bar degrees $\{0,1,2\}$ (Theorem H), so $\mathrm{ChirHoch}^1 = \mathrm{Der}(\mathcal{A})/\mathrm{Inn}(\mathcal{A})$ with no higher corrections. Inner derivations are modes of weight-1 fields acting as $\mathrm{ad}(J_0^a)$; by the Garland–Lepowsky argument ([**?**], adapted to chiral algebras via [**?**]), every weight-0 derivation annihilating the vacuum and commuting with L_{-1} is inner on the subalgebra generated by weight-≥ 2 fields. Hence $\mathrm{ChirHoch}^1(\mathcal{A}) = V_1$ modulo the span of J_0^a-actions that are already inner; for currents at generic level, the Kac–Moody relations make every J^a an independent outer derivation of the non-current subalgebra. □

Specialising: $V_k(\mathfrak{g})$ has $V_1 = \mathfrak{g}$ (currents J^a), giving $\dim \mathrm{ChirHoch}^1 = \dim(\mathfrak{g})$; Vir_c has $V_1 = 0$ (only $L_{-2}\mathbf{1}$ at weight 2), giving $\dim \mathrm{ChirHoch}^1 = 0$; the principal $\mathcal{W}_N$ has primary generators at weights $2, 3, \ldots, N$ with $V_1 = 0$ (the Cartan is frozen by the Drinfeld–Sokolov reduction), giving $\dim \mathrm{ChirHoch}^1(\mathcal{W}_N) = 0$. The degree-0 and degree-2 terms are always present (center and dual center); the degree-1 term is the spin-1 generator count.

10.10 Three features of the census

Three structural features of the census table (Table 1) are visible only across the full landscape.

Shadow depth is not monotone in OPE pole order. The $\beta\gamma$ system has OPE pole order $p_{\max} = 1$ (a single simple pole) yet shadow depth $r_{\max} = 4$, while affine Kac–Moody has

$p_{\max} = 2$ yet $r_{\max} = 3$. Shadow depth is an invariant of the obstruction tower, not of the OPE; the $\beta\gamma$ system separates the two invariants.

The complementarity conductor K is family-specific, not universal. It is 0 for Kac–Moody and free fields, 13 for Virasoro, 250/3 for $\mathcal{W}_3$, 98/3 for Bershadsky–Polyakov. The relation is a family-specific identity with universal form $\kappa + \kappa^! = \varrho_{\mathcal{A}} \cdot K$, where $\varrho_{\mathcal{A}}$ is the per-family leading κ-coefficient ($\varrho_{\mathrm{KM}} = \varrho_{\mathrm{free}} = 0$, $\varrho_{\mathrm{Vir}} = 1/2$, $\varrho_{\mathcal{W}_N} = H_N - 1$, $\varrho_{\mathrm{BP}} = 1/6$); the bare equation $\kappa + \kappa^! = K$ (without ϱ) holds for no standard family.

The R-matrix column compresses to the κ column under averaging. Every entry demonstrates the same phenomenon: $\mathrm{av}(r_{\mathcal{A}}(z)) = \kappa(\mathcal{A})$ (for abelian algebras, directly; for non-abelian Kac–Moody, after accounting for the Sugawara shift). The matrix-valued, z-dependent classical R-matrix of the ordered bar collapses to a single scalar under the Σ_2-coinvariant projection. Every family with $r_{\max} \geq 3$ has an ordered bar that carries strictly more information than the symmetric bar; the five theorems extract the Σ_n-invariant content of this richer structure.

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A Note added in proof: programme strengthenings at the time of first circulation

The 2026-04-16 closure wave and the 2026-04-17 Beilinson audit inscribed a substantial set of strengthenings, reroutings, and scope qualifiers that refine the scope and presentation of the five theorems above without altering their underlying statements. We record them in order of theorem.

(A) Theorem A strengthenings

(A.1) Theorem $\mathrm{A}^{\infty,2}$ **(properad equivalence).** The bar-cobar adjunction of Theorem A lifts to an $(\infty, 2)$-adjoint equivalence at the *properad* level in the Francis–Gaitsgory factorization ambient, inscribed as `thm:A-infinity-2` in Vol I Chapter `chapters/theory/theorem_A_infinity_2.tex`. The pole-free restriction recovers Loday–Vallette's $(\mathrm{Ass}, \mathrm{Ass}^!)$ Koszul pair via $(D_X\text{-mod}, \otimes^!) \hookrightarrow \mathrm{Fact}(X)$; the R-twisted Σ_n-descent lemma `lem:R-twisted-descent` (scoped to $R(z)$ satisfying the classical Yang–Baxter equation and unitarity $R(z)\, R^{op}(-z) =$; automatic for rational Yangians and generic-q trigonometric $U_q(\widehat{\mathfrak{g}})$, vacuous when $R(z) = \tau$) relates $B^{\mathrm{ord}}(\mathcal{A})$ to $B^{\Sigma}(\mathcal{A})$ at properad level and upgrades Theorem A^{E_1} of the E_1 primacy standalone to a corollary of $\mathrm{A}^{\infty,2}$. Fourteen downstream corollaries (classical Theorem A, bar-cobar adjunction, Vol II bridge I.3.2, Vol III Φ, ...) follow.

(A.2) Modular-family Theorem A on $\overline{\mathcal{M}}_{g,n}$ PROVED. The modular-family extension of Theorem A to the full Deligne–Mumford moduli $\overline{\mathcal{M}}_{g,n}$ (including boundary) is now unconditional on the standard landscape. Three independent ingredients converge: (i) hypothesis (c) base-change proved via the Beilinson–Drinfeld holonomic formalism combined with the Vol II six-functor formalism on the relative Ran prestack; (ii) a nodal sewing theorem via Mok25 logarithmic Fulton–MacPherson compactification and Francis–Gaitsgory factorization-gluing, with Huang–Segal sewing convergence carrying chain-level matching at the nodes; (iii) globalization to the full $\overline{\mathcal{M}}_{g,n}$ via the K-theoretic filtration by collision depth, with associated graded equal to Arnold–Orlik–Solomon and the Euler characteristic computed via $\lambda_{-1}(\text{bundle})$ (same template as Theorem D).

(A.3) PTVV/factorization-homology alternative (H01). The factorization-homology alternative proof path for Theorem A, using Costello–Gwilliam factorization homology combined with Pantev–Toën–Vaquié–Vezzosi 1-shifted symplectic Lagrangian correspondence, upgrades from a sketch (H01) to a proposition in the rectified manuscript. The alternative is *genuinely independent*: it invokes neither Francis–Gaitsgory nor the Verdier-pair hypothesis.

(A.4) Six-fold TFAE on the Koszul locus, with a class-G seventh equivalence, at genus 0. The genus-0 clause of Theorem A is sharpened through the hub-and-spoke theorem `thm:ftm-seven-fold-tfae-via-hub-spoke` (Vol I `chapters/theory/ftm_seven_fold_tfae_platonic.tex`): five spokes (Koszul morphism, counit quasi-isomorphism, unit weak equivalence, twisted tensor acyclic, bar concentrated in weight 1) are each bidirected to the PBW E_2-collapse hub on the full Koszul locus, and a sixth spoke (Swiss-cheese formality) is bidirected to the hub on the class-G stratum. The ten transitive bidirections of the six-fold TFAE thus follow from five proved spokes, with a sixth class-G-scoped spoke adjoining the seventh equivalence. Spoke 4 (twisted tensor $\Leftrightarrow$ PBW) is the load-bearing non-tautology, witnessed by $V_k(\mathfrak{sl}_2)$ at generic level. The SC-formality spoke is genuinely class-G scoped: class L/M members are Koszul but not SC-formal. Spoke 5 (bar weight-1 concentration) extends to $g \geq 1$ only for class G; outside class G the fiberwise obstruction $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ is nonzero.

(B) Theorem B scope refinements

(B.1) Weight-completed chiral Positselski inscribed. The coderived bar-cobar adjunction is inscribed as `thm:chiral-positselski-weight-completed` (Vol I `chapters/theory/theorem_B_scope_p` which carries the legitimate phantomsection aliases `thm:chiral-positselski-7-2` and `thm:chiral-positselsk` at the theorem's home. The statement is UNCONDITIONAL on the *weight-completed* bar coalgebra for every chiral algebra with positive-energy conformal-weight grading and finite-dimensional conformal weight spaces. The five-step proof uses the chiral Φ/Ψ adjoints on the weight truncations $C/F^{\leq w}$, the chiral co-contra correspondence `thm:chiral-co-contra-correspondence` at each finite stage, and a Mittag–Leffler Milnor passage to the inverse limit. The chiral adaptations of Positselski [**?**, Theorem 4.6, Theorem 5.3] are invoked at citation level for the inverse-limit compatibility of the coderived and contraderived constructions; the chiral tensor preserves filtered colimits in each variable, making the adaptations immediate.

(B.2) Direct-sum chain-level class M FALSE; weight-completed PROVED. The chain-level class-M statement is genuinely FALSE at the direct-sum level: at generic central charge, $S_4(\mathrm{Vir}_c) = 10/[c(5c+22)] \neq 0$ forces bar cohomology at weight 4, so strict chain-level quasi-isomorphism cannot hold. This is the correct scope, not a gap. The correct chain-level closure is in the *weight-completed* category via the strong filtration $F^n = \bigoplus_{k\leq n} B_k$, established as `thm:completed-bar-cobar-strong` in Vol I `chapters/theory/theorem_B_scope_platonic.tex`. The harmonic discrepancy at each weight stage N is a finite sum absorbed by an explicit chain homotopy $h \cdot m_0^{j-1}$; the Milnor/Mittag-Leffler passage to inverse limit converges.

(B.3) Class G/class L chain-level explicit. Class G Heisenberg chain-level inversion is unconditional via explicit MacLane splitting σ_{Heis} on $\mathrm{Sym}(V_{[1]})$. Class L affine Kac–Moody at non-critical level chain-level inversion is unconditional via the PBW filtration and E_2-collapse lifting.

(C) Theorem C: refinement status

The 2026-04-16 wave proposed nine refinements to Theorem C; the 2026-04-17 honest accounting, after adversarial audit, is recorded at Chapter **??**. The inscribed status is the following.

(C.1) The brace-dg-algebra-level identification $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \cong Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}^!)$ as E_2-algebras after Deligne–Tamarkin is recorded as Conjecture **??**. The H^0-level naive center recovery with the Koszul pairing is proved unconditionally at Lemma **??**.

(C.2) Perfectness on the standard landscape (Proposition **??**) is a conditional proposition in the fiber-finiteness hypothesis. It is unconditional for the Heisenberg family and for affine Kac–Moody at positive integer or boundary admissible level, via Tsuchiya–Ueno–Yamada propagation of finite conformal blocks in the first case and Arakawa C_2-cofiniteness in the second. For affine Kac–Moody at generic non-critical k outside these loci, the boundary-stratum finiteness is recorded as Conjecture **??**. For class M the boundary-stratum fiber-finiteness is recorded as Conjecture **??**.

(C.3) C_1 reflexivity on the Koszul locus is unconditional for the loci of Corollary **??**, and is conditional on Conjecture **??** or Conjecture **??** for the rest of the standard landscape.

(C.4) There is no $+3$ shift contradiction at $g = 0$: the Verdier pairing of degree $-(3g-3)$ belongs to the $g \geq 1$ clause of Theorem **??**; at $g = 0$ the involution is trivial, $Q_0(\mathcal{A}^!) = 0$, and no Verdier pairing is invoked. See Remark **??**.

(C.5) The C_2 hypotheses are pinned: BV package + Verdier + bar-chart lift only. Uniform-weight is *not* a C_2 hypothesis but belongs to the scalar-complementarity corollary.

(C.6) C_0 is unconditional for the two families of Corollary **??**; the extension to class-M standard landscape is conditional on Conjecture **??**.

(C.7) The MC5 strengthening closes class-M C_2(iii) in the weight-completed category.

(C.8) The explicit cross-channel correction at $(W_3, g{=}2)$ is inscribed: $\delta F_2^{\mathrm{cross}}(W_3) = (c + 204)/(16c)$ (Proposition **??**).

(C.9) The mapping-stack $R\mathrm{Map}(\overline{\mathcal{M}}_g, BG_{\mathcal{A}})$ Pantev–Toën–Vaquié–Vezzosi presentation

(Theorem **??**) has mixed status. The shifted-symplectic form on the mapping stack is constructed independently of the eigenspace decomposition; the Lagrangian complementarity clause forwards to Theorem **??** and is tagged `ClaimStatusConjectured` until an intrinsic derived-stack construction of the Verdier anti-symplectic involution is inscribed.

(D) Theorem D: the three prior gaps are closed

All three prior gaps in Theorem D are closed as of 2026-04-16. Part (i), the genus-1 formula $\mathrm{obs}_1 = \kappa \cdot \lambda_1$, remained unconditional throughout.

(D.1) Step 1 virtual-class globalization. Via K-theoretic filtration, $\lambda_{-1}(E)$, and the top-degree Chern-character identity $\mathrm{ch}_g(\lambda_{-1}(E)) = c_g(E)$ (splitting principle).

(D.2) Faltings citation retargeted. The Arakelov–Faltings identification used in Proof A is retargeted from the unpublished Faltings source to Arakelov 1974 plus Faltings 1984 §2 plus Soulé 1992 Ch. III (fiberwise Chern-curvature formula for the Hodge bundle in the Arakelov metric).

(D.3) BGS $c_1(\det E) \to c_g(E)$ bridge. Via the splitting principle combined with scalar-channel linearity: Chern–Weil for a scalar-endomorphism-valued curvature reads off the top Chern class with a *linear* scalar coefficient, not a multiplicative one.

(D.4) Clutching-uniqueness PROVED. The clutching-uniqueness proposition (genus-1 base + separating-clutching additivity + nonseparating-clutching trivialization + tautological-ring purity) pins $\mathrm{obs}_g / \kappa = \lambda_g$ uniquely by the Graber–Vakil socle theorem.

(D.5) H04 upgraded. The factorization-homology + PTVV 1-shifted-symplectic alternative is GENUINELY DISJOINT (derived-categorical, with no uniform-weight hypothesis). This suggests that the uniform-weight restriction of Theorem D(ii) is technical rather than essential.

Consequently Theorem D(ii) is now UNCONDITIONAL on the uniform-weight lane (both routes, Proof A and Proof B of §6.4, stand without a further gap); the scope hedge in Remark 6.2 may be read as historical.

(H) Theorem H: Step 3 circularity rerouted

(H.1) PBW Koszulity criterion. The circularity risk at Step 3 of the $\{0, 1, 2\}$ concentration proof has been rerouted through `thm:pbw-koszulness-criterion` directly, via the Shelton–Yuzvinsky Koszulity of braid-arrangement Orlik–Solomon algebras plus the chiral quadratic-Koszul lemma. Bar-concentration and FM-tower collapse are now *parallel consequences* of the PBW collapse criterion, not sequential, retiring the earlier sequential routing.

(H.2) Three independent-verification decorators. Three disjoint-source verification paths are drafted for Theorem H: Heisenberg via Feigin–Frenkel CE (disjoint from bar); Virasoro via Wang BRST and Feigin–Fuks (disjoint); affine $\mathfrak{sl}_2$ via Whitehead + Künneth (disjoint).

(H.3) E_1-variant provable. The E_1 variant `thm:hochschild-concentration-E1` (for Yangian input) is provable via $\mathrm{FM}^{\mathrm{ord}}$ plus pure-braid Orlik–Solomon plus the ordered Koszul dual; the symmetric-bar obstruction $R(z) \neq \tau$ IS the quantum-group content. The alternative H05 is retrievable as the sharp-Hilbert-series statement from the `fm-tower-collapse` E_2 page.

Structural adjacents

(E1) Iterated-Sugawara tower and E_∞-topologization (Vol II). `thm:iterated-sugawara-construction` and `thm:e-infinity-topologization-ladder` (Vol II `chapters/connections/e_infinity_topologization.` promote the Sugawara construction of a single stress tensor to an infinite tower of inner higher-spin Casimir tensors, each of which raises the topologization level by one: k inner stress tensors yield E_{k+2}^{top}. Virasoro at non-critical level gives E_3^{top}; $\mathcal{W}_N$ gives E_{N+1}^{top}; $\mathcal{W}_\infty$ gives E_∞^{top} as the Platonic endpoint. This closes the prior open frontier "chain-level topologization on the original complex for class M" via the weight-completed route plus the half-BRST pattern.

(E2) Chiral higher Deligne (Vol II). `thm:chiral-higher-deligne` (Vol II `chapters/theory/chiral_hi` identifies $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ as the universal E_3-chiral algebra acted on by $\mathrm{SC}^{\mathrm{ch,top}}$ via heptagon edges (3) $\leftrightarrow$ (4) $\leftrightarrow$ (5). Theorem H concentration is thereby a CONSEQUENCE of E_3-rigidity-at-a-point plus PBW collapse.

(E3) Periodic CDG for admissible KL (Vol I). `thm:periodic-cdg-is-koszul-compatible` (Vol I `chapters/theory/periodic_cdg_admissible.tex`) closes the admissible-level Kazhdan–Lusztig case: at $k + h^\vee = p/q$, the periodic-CDG filtration $F^n = \ker(Q^n_{\mathrm{adm}})$ on $\mathrm{KL}^{\mathrm{adm}}_k$ is compatible with chiral Koszul duality; the chiral adjunction $\Omega^{\mathrm{ch}} \dashv B^{\mathrm{ch}}$ descends UNCONDITIONALLY.

Beilinson-rectified open frontiers (2026-04-17)

The 2026-04-17 Beilinson audit surfaced the following scope qualifiers and retractions that refine, but do not invalidate, the five theorems.

(F.1) $\mathcal{W}(p)$ triplet tempering retracted. The Vol II inscription of `thm:tempered-stratum-contains-wp` is RETRACTED from ProvedHere to Conjectured. The Zhu-bounded-Massey proof chain fails: Gurarie 1993 and Flohr 1996 construct logarithmic-CFT amplitudes with unbounded Massey products despite finite-dimensional Zhu algebra. The corrected tempered scope is: principal, non-logarithmic, non-minimal standard landscape; $\mathcal{W}(p)$ is open pending an Adamović–Milas character-amplitude bound.

(F.2) Programme-climax emptiness SCOPE-QUALIFIED. The programme-climax statement "the non-tempered stratum is empty on the C_2-cofinite standard landscape" is scope-qualified: emptiness holds on the non-logarithmic subset (Virasoro, $\mathcal{W}_N$, Schellekens, Monster, irrational cosets); logarithmic $\mathcal{W}(p)$ remains open.

(F.3) Kummer-irregular prime labels retracted. Primes $1423, 3067, 23, 43, 419$ are RETRACTED from the Kummer-irregular label: primary-source Bernoulli witness search yields no witness. They still appear in S_r numerators as Riccati-arithmetic characteristic primes, NOT Kummer-arithmetic. The corrected Tier-3 emergence set is $\{37, 691, 811\}$.

(F.4) Super-Yangian complementarity. The super-Yangian complementarity sum SPLITS by normalisation. The super-trace form $\kappa + \kappa^{,!} = 0$ is PROVED for $Y_\hbar(\mathfrak{sl}(m|n))$ on the sub-Sugawara line $m \neq n$ via super-Sugawara rational-identity cancellation (`prop:super-berezinian-central-` in `yangians_foundations.tex`). The Berezinian form $\kappa + \kappa^{,!} = \max(m, n)$ at the Sugawara-shifted dual level is CONJECTURED (`conj:super-berezinian-shadow-shift-magnitude`), conditional on a closed-form derivation of the $\frac{1}{2}\max(m, n)$ shift magnitude induced by the central automorphism σ. Nazarov 1991 and Gow 2006 supply centrality of $(T(u))$ and non-degeneracy of the Berezinian pairing, not the shift magnitude in closed form; the identity is verified symbolically at small rank as evidence, not derivation.

(F.5) CY-C pentagon stratifies generator rank, not κ_{ch} (Vol III). The Vol III pentagon of five intertwiners stratifies the generator-lattice rank $\rho^{R_i} \in \{3, 12, 24\}$, NOT the chiral modular characteristic. κ_{ch} is route-independent and equal to zero for K3×E by the Hodge supertrace. This is a CY-side correction with no effect on the five theorems of Vol I at their stated scopes.

Table 1: Invariants of the standard landscape. All R-matrices in the trace-form convention; Ω is the inverse Killing form Casimir; $d = \dim \mathfrak{g}$, $t = k + h^\vee$.

Family	c	κ	K	Δ	**Class**	$r_{\max}$	R**-matrix**
Abelian (class G, Gaussian)							
$\mathcal{H}_k$ (Heisenberg)	1	k	0	0	G	2	k/z
Free fermion ψ	1/2	1/4	0	0	G	2	$\frac{1}{2z}$
Lattice V_Λ	rank(Λ)	rank(Λ)	0	0	G	2	Heisenberg
Affine Kac–Moody (class L, Lie)							
$\widehat{\mathfrak{sl}}_2$ at level k	$\dfrac{3k}{k+2}$	$\dfrac{3(k+2)}{4}$	0	0	L	3	$k\,\Omega/z$
$\widehat{\mathfrak{sl}}_3$ at level k	$\dfrac{8k}{k+3}$	$\dfrac{4(k+3)}{3}$	0	0	L	3	$k\,\Omega/z$
$\widehat{\mathfrak{so}}_8$ (D_4)	$\dfrac{28k}{k+6}$	$\dfrac{7(k+6)}{3}$	0	0	L	3	$k\,\Omega/z$
$\widehat{G}_2$	$\dfrac{14k}{k+4}$	$\dfrac{7(k+4)}{4}$	0	0	L	3	$k\,\Omega/z$
$\widehat{F}_4$	$\dfrac{52k}{k+9}$	$\dfrac{26(k+9)}{9}$	0	0	L	3	$k\,\Omega/z$
$\widehat{E}_6$	$\dfrac{78k}{k+12}$	$\dfrac{13(k+12)}{4}$	0	0	L	3	$k\,\Omega/z$
$\widehat{E}_7$	$\dfrac{133k}{k+18}$	$\dfrac{133(k+18)}{36}$	0	0	L	3	$k\,\Omega/z$
$\widehat{E}_8$	$\dfrac{248k}{k+30}$	$\dfrac{62(k+30)}{15}$	0	0	L	3	$k\,\Omega/z$
Free fields (class C/G)							
$\beta\gamma$ (weight λ)	$2(6\lambda^2 - 6\lambda + 1)$	1	0	special	C	4	constant
bc ghosts (weight λ)	$1 - 3(2\lambda - 1)^2$	$c_{bc}/2$	0	0	G	2	Clifford
$\mathcal{W}$-algebras and Virasoro (class M, Mixed)							
Vir_c	c	$c/2$	13	$\neq 0$	M	∞	$\dfrac{c/2}{z^3} + \dfrac{2T}{z}$
$\mathcal{W}_3$ ($\mathfrak{sl}_3$ DS)	c	$5c/6$	250/3	$\neq 0$	M	∞	matrix-valued
$\mathcal{W}_N$ ($\mathfrak{sl}_N$ DS)	c	$c(H_N - 1)$	N-dep.	$\neq 0$	M	∞	matrix-valued
Non-principal $\mathcal{W}$-algebras							
BP_k (Bersh.–Pol.)	$2 - \dfrac{24(k+1)^2}{k+3}$	$c/6$	98/3	$\neq 0$	M	∞	multi-channel